

Flow-Tek®

A Subsidiary of BRAY INTERNATIONAL, Inc.

Trunnion Ball Valve

NPS 2" to 56"; DN 50 to 1400
Up to ASME Class 2500
Cast & Forged Construction





INTRODUCTION

Petro-seal series valves are trunnion mounted, polymeric-seated ball valves, which provide excellent tightness over an extensive range of temperatures and pressures. They are available in both full bore and reduced bore designs in a choice of materials, sizes and pressure classes which makes them an ideal selection for a variety of application.

This extensive range of sizes, styles, pressure classes and materials together with conformance to ANSI, API, and NACE specifications qualify these valves for virtually all oil and gas services.

A trunnion ball valve has additional mechanical anchoring of the ball at the top and the bottom, suitable for larger and higher pressure services. To ensure tight sealing at low pressure, high-tensile springs force the seats against the ball while at higher pressure, the medium pressure pushes upstream seat towards the ball.

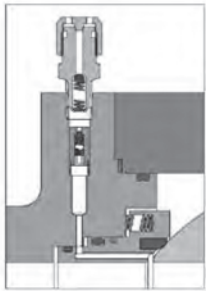
TRUNNION BALL VALVE

INTRODUCTION

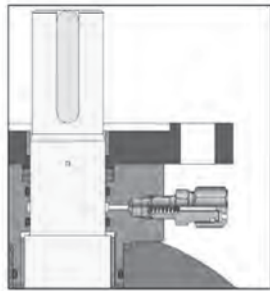
FEATURES

EMERGENCY SEALS

For trunnion mounted ball valve size NPS 6 (DN 150) and above, it comes with provisions for sealant injection on both the stem and seat while for sizes NPS 5 (DN 125) and below on body cavity. In case of failed underperformed seals, a temporary emergency seal can be achieved by injecting sealants.



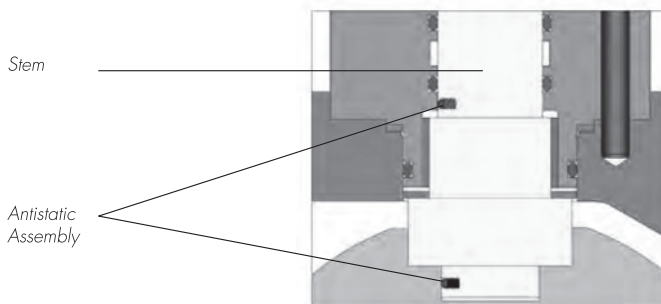
Seat Sealant Valve



Stem Sealant Valve

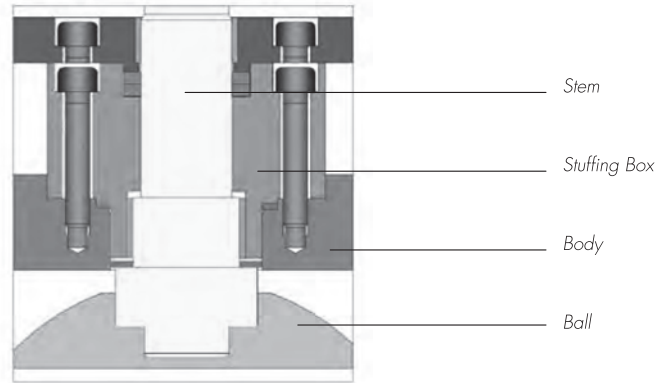
ANTI-STATIC

Because the ball and stem in a ball valve are suspended on non-metallic parts, i.e. the seat seal and stem seal, there is a possibility a static charge may build up on the stem-ball, a mechanical (antistatic metal spring and ball) is introduced in the design to maintain the metal- to-metal contact between the rotating ball/stem and the valve body which will ground any charges to the valve body.



BLOWOUT PROOF STEM

The valve stem is made with a shoulder at the bottom end. It's securely retained by the stuffing box, to avoid that the stem, under certain operating conditions, accidentally blows out. Other designs are available on request.



CORROSION RESISTANCE

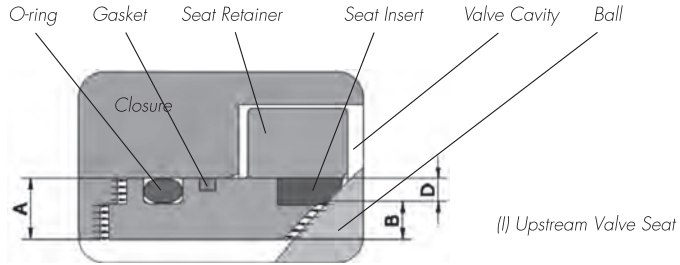
Carbon Steel stems, balls and seats are Electroless Nickel Plated (ENP) as standard. A wide range of alloys are available for corrosive services.

NACE COMPLIANCE

FlowTek trunnion ball valves are supplied as standard in materials that meet the requirements of NACE MR 01-75. Strict material control and inspection during the manufacturing process insure compliance.



SINGLE PISTON EFFECT

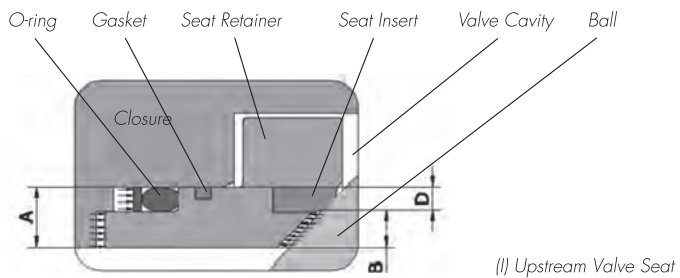


The difference in area (D) times the line pressure creates a "piston effect" force which pushes the seat against the ball surface resulting in a tight effect seal.

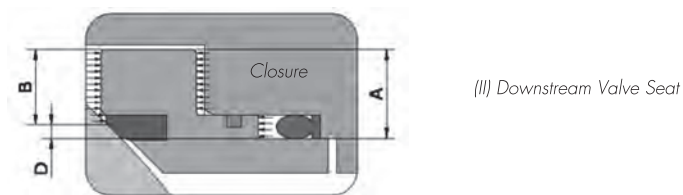


When the pressure in the valve cavity reaches a level that the total force from cavity acting on the seat is larger than the total force from upstream line pressure, the seat will be pushed away from the ball to relief the valve cavity pressure.

DOUBLE PISTON EFFECT



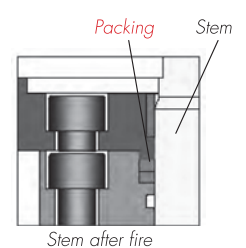
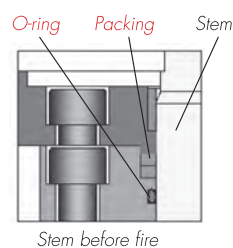
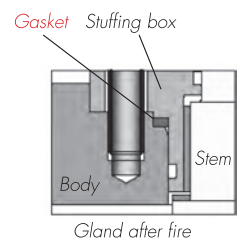
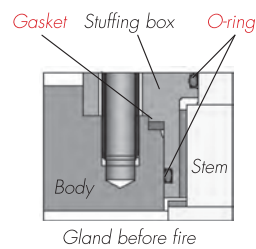
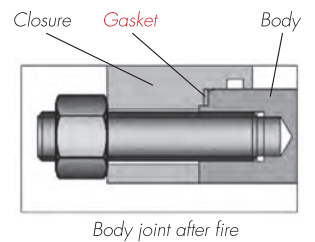
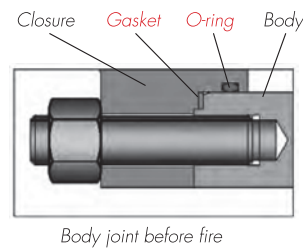
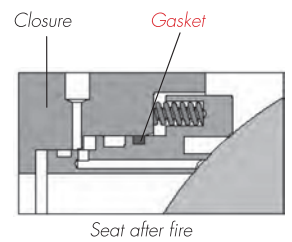
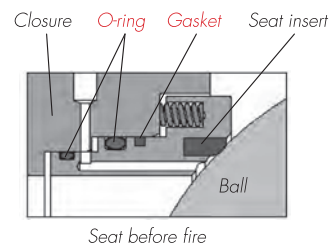
The different in area (D) times the line pressure creates a "piston effect" force which pushes the seat against the ball surface resulting in a tight seal.



The difference in the area (D) times the cavity pressure creates a "piston effect" force which pushes the seat against the ball surface resulting in a tight seal.

FIRE SAFE DESIGN

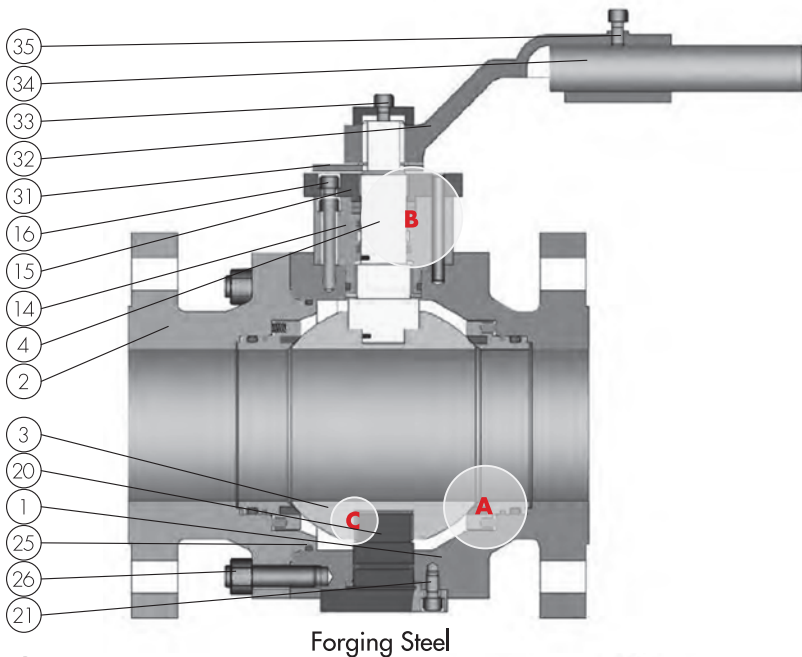
During a fire, non-metallic soft material will be burnt, subsequently seat leakage or external leakage may occur and cause the fire spread or contaminate the environment. Flow-Tek ball valves are fire tested in accordance with API 6FA or API 607.



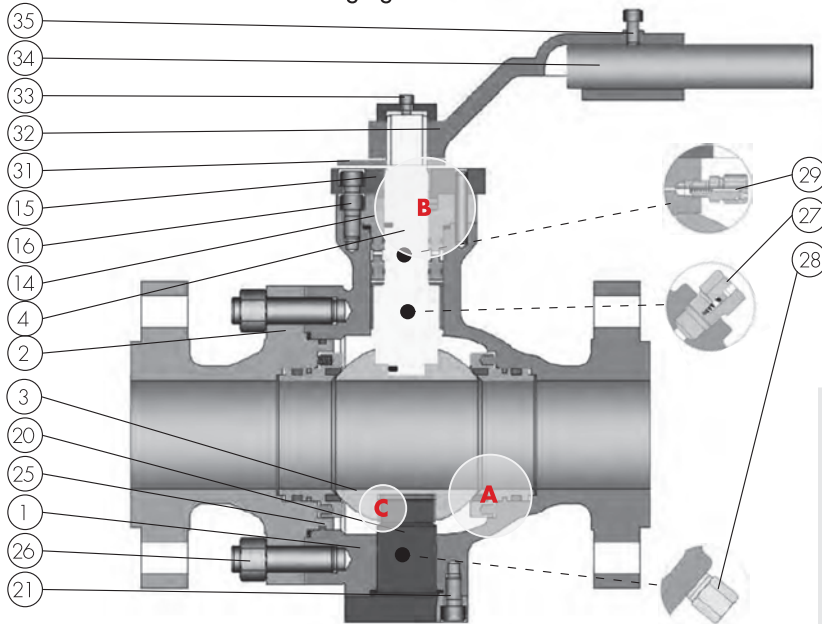
TRUNNION BALL VALVE

OVERVIEW

2-PIECE SPLIT BODY



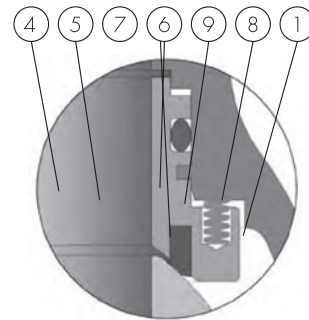
Forging Steel



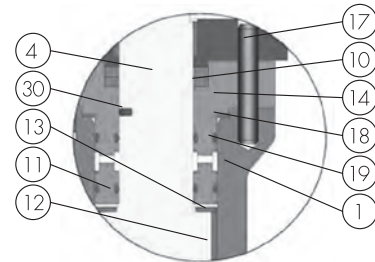
Casting Steel

PART LIST

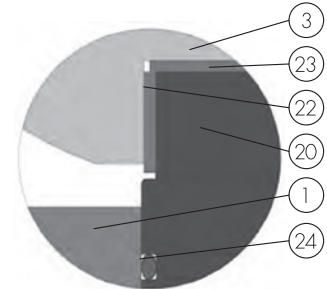
- | | | |
|-----------------|--------------------------|------------------------|
| 1 Body | 13 Thrust washer | 25 Body seal |
| 2 Closures | 14 Stuffing box | 26 Stud bolt & nut |
| 3 Ball | 15 Gland cover | 27 Vent valve |
| 4 Stem | 16 Gland socket bolt | 28 Drain plug |
| 5 Seat insert | 17 Gland pin | 29 Stem sealant valve |
| 6 Seat retainer | 18 Gland seal | 30 Antistatic assembly |
| 7 Seat spring | 19 Gland O-ring | 31 Stopper |
| 8 Seat O-ring | 20 Trunion | 32 Lever |
| 9 Seat seal | 21 Trunion socket bolt | 33 Lever lock screw |
| 10 Stem packing | 22 Trunion bushing | 34 Extension pipe |
| 11 Stem O-ring | 23 Trunion thrust washer | 35 Pipe lock screw |
| 12 Stem bushing | 24 Trunion O-ring | |



A



B



C

STANDARDS

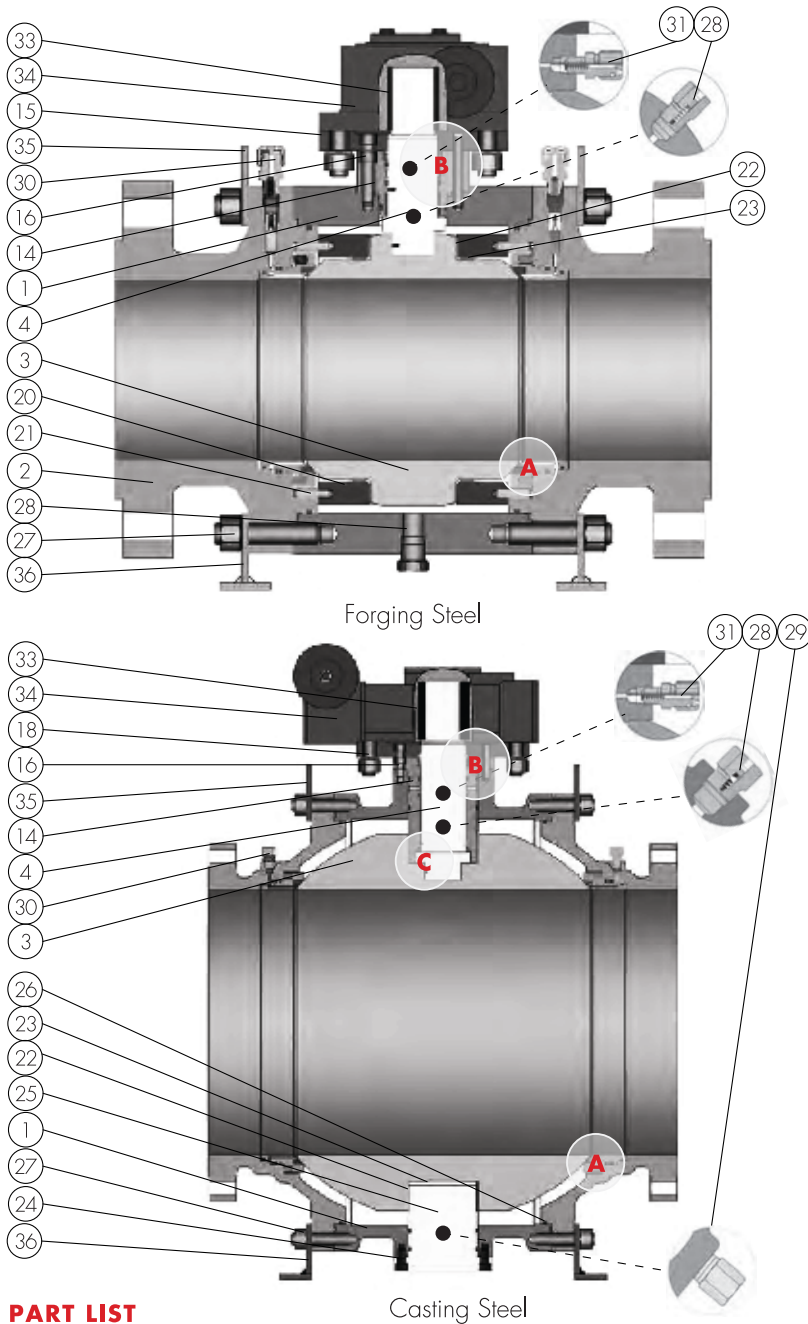
Design & manufacture	API 6D, ISO 14313, ISO 17292
Face-to-face	API 6D, ASME B16.10
End flanges	ASME B16.5, ASME B16.47 MSS SP-44 (NPS 22 Only)
Test & inspection	API 6D, API 598
Fire safe	API 6FA, API 607
Other	NACE MR 01-75, MR 0103 ISO 15848, API 622, API 624

TYPICAL MATERIALS

Body/Closures	A105, A182 F304, F304L, F316, F316L, F51, F53, A350 LF2, LF3, LF5 A216 WCB, A351 CF3, CF8, CF3M, CF8M, A995 4A, 5A, A352 LCB, LCC, LC2 (Casting)
Ball	CS+ENP, A182 F304, F304L, F316, F316L, F51, F53, CS+TCC, CS+Ni60
Seat retainer	CS+ENP, A182 F304, F304L, F316, F316L, F51, F53, CS+TCC, CS+Ni55
Seat insert	PTFE, RPTFE, Nylon, Devlon, PEEK
Stem	A182 F6a, F316, F51, A105+ENP, AISI 4140+ENP, 17-4PH
Packing	Graphite, PTFE, RPTFE
O-ring	Viton, NBR, HNBR, AFLAS

TRUNNION BALL VALVE

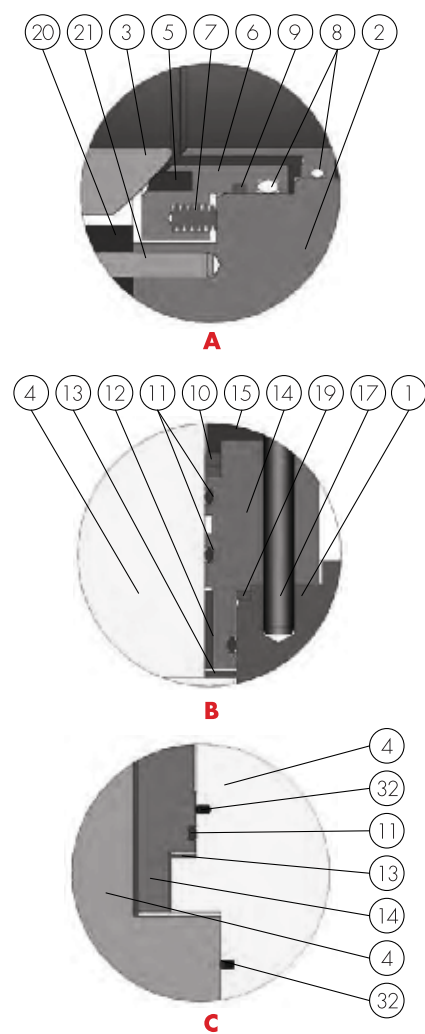
OVERVIEW



PART LIST

1 Body	13 Thrust washer	25 Trunnion
2 Closures	14 Stuffing box	26 Body seal
3 Ball	15 Gland cover	27 Stud bolt & nut
4 Stem	16 Gland socket bolt	28 Vent valve
5 Seat insert	17 Gland pin	29 Drain plug
6 Seat retainer	18 Gland bolt & nut	30 Seat sealant valve
7 Seat spring	19 Gland seal	31 Stem sealant valve
8 Seat O-ring	20 Trunnion support	32 Antistatic assembly
9 Seat seal	21 Trunnion support pin	33 Drive key
10 Stem packing	22 Bushing	34 Gearbox assembly
11 Stem O-ring	23 Thrust washer	35 Lifting lug
12 Stem bushing	24 Trunnion socket bolt	36 Supporting leg

3-PIECE SPLIT BODY



STANDARDS

Design & manufacture	API 6D, ISO 14313, ISO 17292
Face-to-face	API 6D, ASME B16.10
End flanges	ASME B16.5, ASME B16.47 MSS SP-44 (NPS 22 Only)
Test & inspection	API 6D, API 598
Fire safe	API 6FA, API 607
Other	NACE MR 01-75, MR 0103 ISO 15848, API 622, API 624

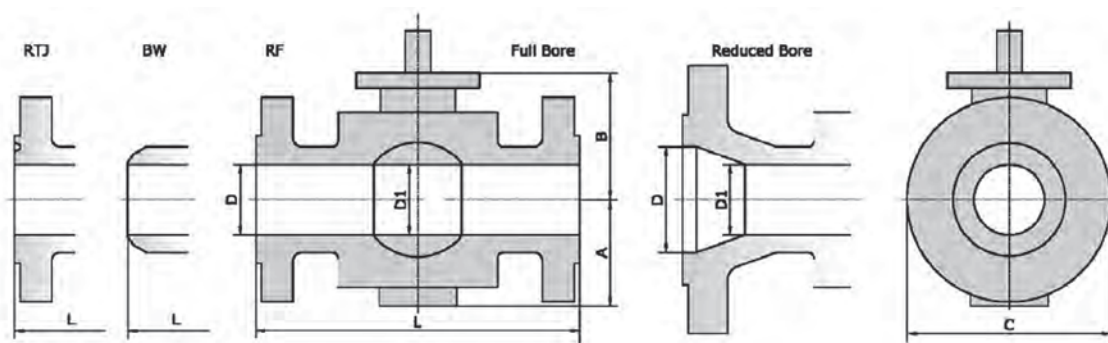
TYPICAL MATERIALS

Body/Closures	A105, A182 F304, F304L, F316, F316L, F51, F53, A350 LF2, LF3, LF5 (Forging) A216 WCB, A351 CF3, CF8, CF3M, CF8M, A995 4A, 5A, A352 LCB, LCC, LC2 (Casting)
Ball	CS+ENP, A182 F304, F304L, F316, F316L, F51, F53, CS+TCC, CS+Ni60
Seat retainer	CS+ENP, A182 F304, F304L, F316, F316L, F51, F53, CS+TCC, CS+Ni55
Seat insert	PTFE, RPTFE, Nylon, Devlon, PEEK
Stem	A182 F6a, F316, F51, A105+ENP, AISI 4140+ENP, 17-4PH
Packing	Graphite, PTFE, RPTFE
O-ring	Viton, NBR, HNBR, AFLAS

TRUNNION BALL VALVE

DIMENSIONS AND WEIGHTS

ASME CLASS 150 (PN 20)



ASME CLASS 150 (PN 20) FULL BORE

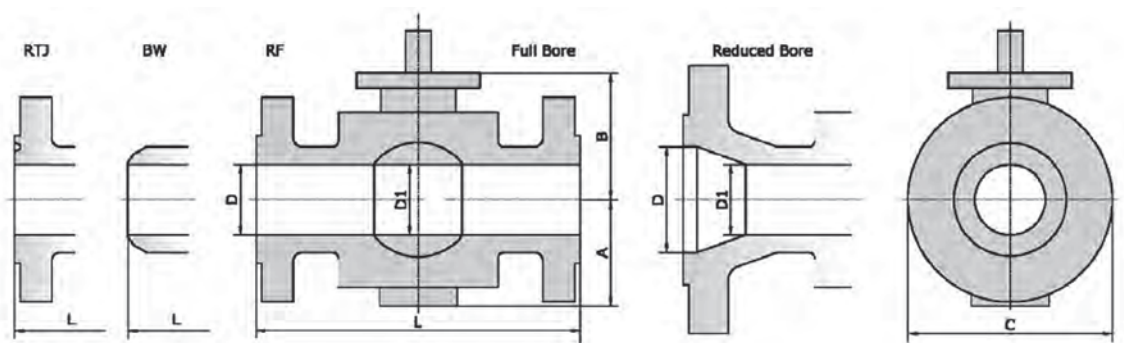
Size in/mm	D	D1	L			Casting			Forging			Weight	
			RF	RTJ	BVV	A	B	C	A	B	C	RF/RTJ (lb/kg)	BVV (lb/kg)
2	1.93	1.93	7	7.5	8.5	3.4	5.3	5.9	3.1	4.6	5.9	40	35
50	49	49	178	191	216	86	135	150	79	117	150	18	16
2 1/2	2.44	2.44	7.5	8	9.5	3.9	5.9	7.3	3.6	5.7	7.1	62	55
65	62	62	191	203	241	100	150	185	92	145	180	28	25
3	2.91	2.91	8	8.5	11	4.9	6.3	7.8	4	5.6	7.7	73	66
80	74	74	203	216	283	125	161	198	102	141	195	33	30
4	3.94	3.94	9	9.5	12	5.2	7.5	9	4.8	6.7	9	110	99
100	100	100	229	242	305	132	190	230	123	170	230	50	45
5	5	5	14	14.53	14	6.1	8.1	11.2	5.9	7.9	11	238	224
125	127	127	356	369	356	155	205	285	150	200	280	108	102
6	5.91	5.91	15	16	18	7.2	9.2	12.6	6.6	8.7	12.1	286	271
150	150	150	394	407	457	184	233	320	168	222	308	130	123
8	7.91	7.91	18	18.5	20.5	9	10	15.7	9	10	15.3	506	484
200	201	201	457	470	521	228	253	398	230	255	388	230	220
10	9.92	9.92	21	21.5	22	10.6	11.4	19.1	11.2	11.8	18.7	847	770
250	252	252	533	546	559	268	290	485	285	300	475	385	350
12	11.94	11.94	24	24.5	25	12.9	15.6	22.2	13.4	14.4	21.4	1,342	1232
300	303	303	610	623	635	327	395	565	340	340	543	610	560
14	13.15	13.15	27	27.5	30	14.8	15.6	14	14.2	15	23.5	1760	1703
350	334	334	686	699	762	375	395	610	360	381	598	800	774
16	15.16	15.16	30	30.5	33	16	17.8	27.9	15.7	16.5	26.7	2640	2420
400	385	385	762	775	838	405	452	708	400	419	678	1200	1100
18	17.17	17.17	34	34.5	36	18.3	18.5	30.1	18.1	18.5	29.9	3388	3124
450	436	436	864	877	914	465	470	765	460	470	760	1540	1420
20	19.17	19.17	36	36.5	39	21.7	21.7	34.3	19.3	19.9	33.1	4048	3850
500	487	487	914	927	991	550	550	870	490	505	840	1840	1750
22	21.19	21.19	39	39.5	43	19.1	19.1	37	18.9	21.5	36.8	4928	4774
550	538	538	991	1003	1092	485	550	940	480	545	935	2240	2170
24	23.19	23.19	42	42.5	46	24	24.1	40.4	23.6	23.9	39.8	6820	6006
600	589	589	1067	1080	1143	610	612	1026	600	607	1010	3100	2730
26	24.94	24.94	45	(1)	49	24.8	26	43.7	24.5	25.6	42.5	8030	7524
650	633	633	1143	(1)	1245	630	660	1110	623	651	1080	3650	3420
28	26.94	26.94	49	(1)	53	27	28	46.3	26.8	27.2	45.6	9636	9064
700	684	684	1245	(1)	1346	685	700	1175	680	690	1157	4380	4120
30	28.94	28.94	51	(1)	55	28	30.6	49.8	29	29	48.8	11814	10780
750	735	735	1295	(1)	1397	710	777	1265	737	737	1240	5370	4900
32	30.69	30.69	54	(1)	60	30.3	30.7	45.1	30.1	30.5	52.6	13310	12716
800	779	779	1372	(1)	1524	770	780	1145	765	775	1335	6050	5780
34	32.69	32.69	58	(1)	64	31.3	30.9	45.3	30.7	30.3	44.5	15180	14146
850	830	830	1473	(1)	1626	795	785	1150	780	770	1130	6900	6430
36	34.41	34.41	60	(1)	68	35.6	34.4	58.3	33.5	33.5	57.8	18920	16720
900	874	874	1524	(1)	1727	905	875	1480	850	852	1468	8600	7600
40	38.44	38.44	70.87	(1)	70.87	37	37	66.9	36.6	36.6	64.2	24200	21340
1000	976	976	1800	(1)	1800	940	940	1700	930	930	1630	11000	9700
42	40.19	40.19	74.8	(1)	74.8	38.4	38.4	66.9	37.4	37.94	66.5	31900	24816
1050	1010	1010	1900	(1)	1900	975	975	1700	950	950	1690	14500	11280
48	45.94	45.94	82.68	(1)	82.68	42.7	45.3	76.4	42.3	43.3	76	39600	35420
1200	1166	1166	2100	(1)	2100	1085	1150	1940	1075	1100	1930	18000	16100
56	53.56	53.56	94.49	(1)	94.49	50.4	51.6	90.2	50	51.6	89.6	61600	55660
1400	1360	1360	2400	(1)	2400	1280	1310	2290	1270	1310	2275	28000	25300

(1) Contact our sales representative for more information.

TRUNNION BALL VALVE

DIMENSIONS AND WEIGHTS

ASME CLASS 150 (PN 20)



ASME CLASS 150 (PN 20) REDUCED BORE

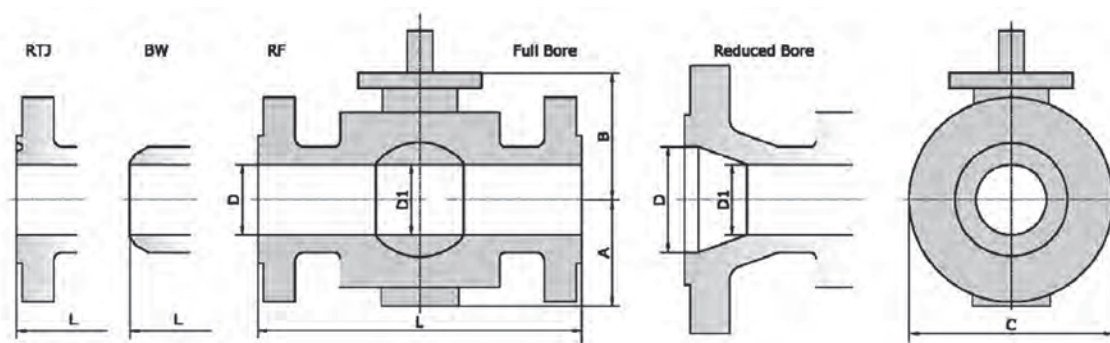
Size in/mm	D	D1	L			Casting			Forging			Weight	
			RF	RTJ	BVV	A	B	C	A	B	C	RF/RTJ (lb/kg)	BVV (lb/kg)
2x1 1/2	1.94	1.5	7	7.5	8.5	3.3	4.9	5.9	3	4.3	5.9	33	26
50x40	49	38	178	191	216	85	125	150	75	110	150	15	12
2 1/2x2	2.44	1.94	7.5	8	9.5	3.3	5.3	7.1	3.1	4.6	7.1	49	42
65x50	62	49	190	203	241	86	135	180	79	117	180	22	19
3x2	2.94	1.94	8	8.5	11.13	3.3	5.3	7.5	3.1	4.6	7.5	55	48
80x50	74	49	203	216	283	86	135	190	79	117	190	25	22
4x3	3.94	2.94	9	9.5	12	4.9	9.1	7.8	4	5.5	9.1	97	81
100x80	100	74	229	241	305	125	230	198	102	141	230	44	37
6x4	5.94	3.94	15.5	16	18	5.5	7.5	11	4.8	6.7	11	154	147
150x100	150	100	394	406	457	140	190	280	123	170	280	70	67
8x6	7.94	5.94	18	18.5	20.5	7.2	9.2	13.6	6.8	8.7	13.6	436	414
200x150	201	150	457	470	521	184	233	345	172.5	222	345	198	188
10x8	9.94	7.94	21	21.5	22	9	10	1.6	9.1	10	15.9	715	671
250x200	252	201	533	546	559	228	253	405	230	255	405	325	305
12x10	11.94	9.94	24	24.5	25	10.6	11.4	19.1	11.2	11.8	19.1	913	843
300x250	303	252	610	622	635	268	290	485	285	300	485	415	383
14x10	13.19	9.94	27	27.5	30	10.6	11.4	21	11.2	11.8	21	1151	1104
350x250	334	252	686	699	762	268	290	535	285	300	535	523	502
16x12	15.19	11.94	30	30.5	33	12.9	15.6	22.2	13.4	13.4	23.4	1639	1551
400x300	385	303	762	775	838	327	395	565	340	340	595	745	705
18x14	17.19	13.19	34	34.5	36	14.8	15.6	25	14.2	15	25	2288	2002
450x350	436	334	864	877	914	375	395	635	360	381	635	1040	910
20x16	19.19	15.19	36	36.5	39	15.9	17.8	27.9	15.7	21.7	27.6	2574	2134
500x400	487	385	914	927	991	405	452	708	400	419	700	1170	970
24x20	23.19	19.19	42	42.5	45	21.7	21.7	34.3	19.3	19.9	33.1	4213	4026
600x500	589	487	1067	1080	1143	550	550	870	490	505	840	1915	1830
30x24	28.94	23.19	51		55	24	24.1	40.4	23.6	23.9	39.8	2948	7744
750x600	735	589	1295	(1)	1397	610	612	1026	600	607	1010	1340	3520
36x30	34.44	28.94	60		68	28	30.6	49.8	29	29	48.8	13420	12760
900x750	874	735	1524	(1)	1727	710	777	1265	737	737	1240	6100	5800

(1) Contact our sales representative for more information.

TRUNNION BALL VALVE

DIMENSIONS AND WEIGHTS

ASME CLASS 300 (PN 50)



ASME CLASS 300 (PN 50) FULL BORE

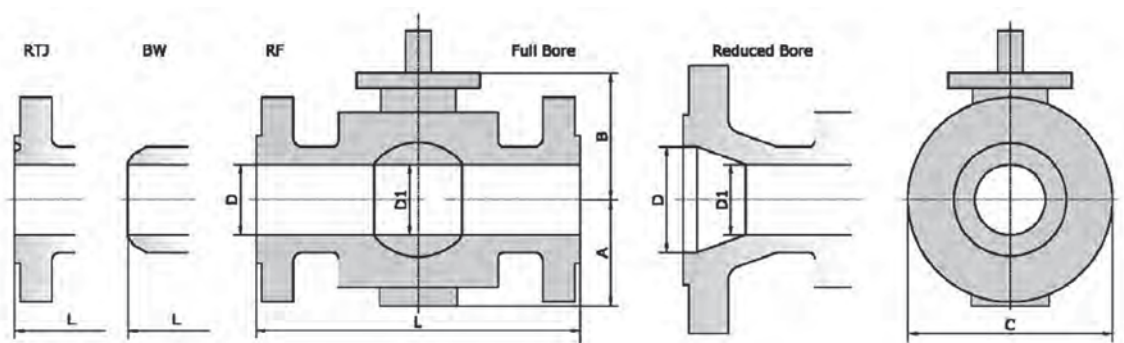
Size in/mm	D	D1	L			Casting			Forging			Weight	
			RF	RTJ	BVV	A	B	C	A	B	C	RF/RTJ (lb/kg)	BVV (lb/kg)
2	1.93	1.93	8.5	9.13	8.5	3.6	5.3	6.5	3.3	4.6	6.5	62	51
50	49	49	216	232	216	91	135	165	83	117	165	28	23
2 1/2	2.44	2.44	9.5	10.12	9.5	3.7	5.9	7.3	3.6	5.7	7.1	99	92
65	62	62	241	257	241	95	150	185	92	145	180	45	42
3	2.94	2.94	11.13	11.75	11.13	5	6.3	8.3	4.1	5.6	8.3	123	92
80	74	74	282	298	283	126	161	210	105	141	210	56	42
4	3.94	3.94	12	12.63	12	5.2	7.5	10	5	6.7	10	172	132
100	100	100	305	321	305	132	190	255	128	170	255	78	60
5	5	5	15.75	16.26	15.75	6.3	8.3	11.6	6.1	8	11.2	328	304
125	125	125	400	413	400	160	210	295	155	202	285	149	138
6	5.94	5.94	15.88	16.5	18	7.4	9.2	13	6.6	8.7	12.6	396	308
150	150	150	403	419	457	189	233	330	168	222	320	180	140
8	7.91	7.91	19.75	20.38	20.5	9	10	15.7	9.3	10	15.4	647	539
200	201	201	502	518	521	228	253	398	235	255	392	294	245
10	9.92	9.92	22.38	23	22	10.6	11.4	10.1	11.4	12	19.1	869	726
250	252	252	568	584	559	268	290	485	290	305	485	395	330
12	11.94	11.94	25.5	28.13	25.5	12.9	15.6	23.2	13.4	13.4	21.8	1463	1232
300	303	303	648	664	635	327	395	590	340	340	554	665	560
14	13.19	13.19	30	30.62	30	15.2	15.7	24.8	14.5	15	23.9	2178	1892
350	334	334	762	778	762	385	400	630	368	384	606	990	860
16	15.19	15.19	33	33.63	33	15.9	17.8	28.2	16.5	16.5	27.1	2829	2277
400	385	385	838	854	838	405	452	716	420	419	689	1286	1035
18	17.19	17.19	36	36.63	36	18.9	19.3	31.5	18.5	18.9	30.7	3630	2860
450	436	436	914	930	914	480	490	800	470	480	780	1650	1300
20	19.17	19.17	39	39.75	39	21.7	21.7	34.3	19.5	19.9	33.5	4268	3872
500	487	487	991	1010	991	550	550	870	495	505	850	1940	1760
22	21.19	21.19	43	43.88	43	20.3	21.7	37.8	20.1	21.5	37.2	5390	4840
550	538	538	1092	1114	1092	515	550	960	510	545	945	2450	2200
24	23.19	23.19	45	45.888	45	24	24.8	40.6	24.8	23.9	40.9	6622	6336
600	589	589	1143	1165	1143	610	630	1030	630	607	1038	3010	2880
26	24.94	24.94	49	50	49	28.4	26	43.1	24.5	25.6	42.7	8404	7810
650	633	633	1245	1270	1245	630	660	1095	623	651	1085	3820	3550
28	26.94	26.94	53	54	53	27.6	28	46.9	27.2	27.6	46.3	10604	9372
700	684	684	1346	1372	1346	700	710	1190	690	700	1175	4820	4260
30	28.94	28.94	55	56	55	28	30.6	49.8	29.4	29.4	49.8	12320	11330
750	735	735	1397	1422	1397	710	777	1265	747	747	1266	5600	5150
32	30.69	30.69	60	61.13	60	30.7	31.1	53.1	30.3	30.7	53	14190	12430
800	779	779	1524	1552	1524	780	790	1350	770	780	1345	6450	5650
34	32.69	32.69	64	65.1	64	31.9	30.5	47.8	31.9	30.5	47.8	15510	14190
850	830	830	1626	1654	1626	810	775	1215	810	775	1215	7050	6450
36	34.41	34.41	68	69.13	68	35.6	34.4	58.3	34	34	58.7	19800	18480
900	874	874	1727	1756	1727	905	875	1480	864	864	1492	9000	8400
40	38.44	38.44	76.77	(1)	76.77	37	37	65	36.6	36.6	64.6	24200	22660
1000	976	976	1950	(1)	1950	940	940	1650	930	930	1640	11000	10300
42	40.19	40.19	82.68	(1)	82.68	38.2	38.2	66.9	37.8	37.8	66.5	31900	29920
1050	1020	1020	2100	(1)	2100	970	970	1700	960	960	1690	14500	13600
48	45.94	45.94	85.43	(1)	85.43	44.1	44.1	77.4	43.3	43.3	76.6	39600	37620
1200	1166	1166	2170	(1)	1270	1120	1120	1965	1100	1100	1945	18000	17100
56	53.56	53.56	98.43	(1)	98.43	50.4	53.1	90.4	50	52.2	89.8	60060	56760
1400	1360	1360	2500	(1)	2500	1280	1350	2295	1270	1325	2282	27300	25800

(1) Contact our sales representative for more information.

TRUNNION BALL VALVE

DIMENSIONS AND WEIGHTS

ASME CLASS 300 (PN 50)



ASME CLASS 300 (PN 50) REDUCED BORE

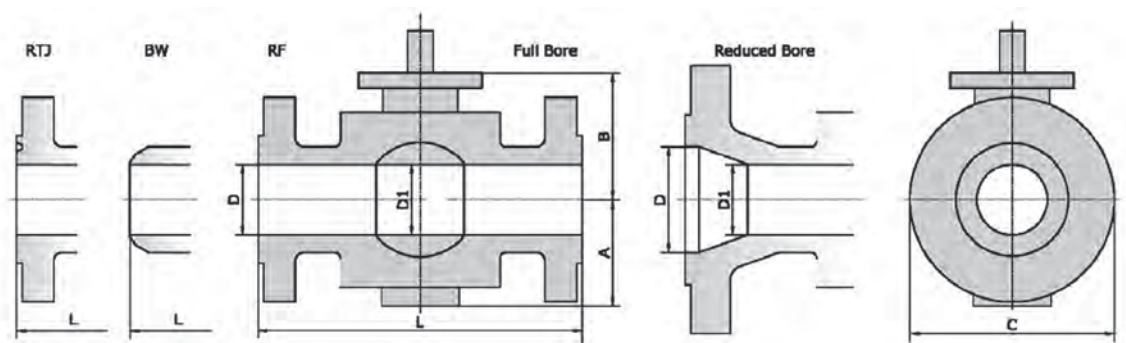
Size in/mm	D	D1	L			Casting			Forging			Weight	
			RF	RTJ	BW	A	B	C	A	B	C	RF/RTJ (lb/kg)	BW (lb/kg)
2x1 1/2 50x40	1.94 49	1.5 38	8.5 216	9.11 232	8.5 216	3.5 90	5.1 130	6.5 165	3.3 83	4.3 110	6.5 165	40 18	26 12
2 1/2x2 65x50	2.44 62	1.94 49	9.5 241	10.13 257	9.5 241	3.6 91	5.3 135	6.5 165	3.3 83	4.6 117	7.5 190	57 26	48 22
3x2 80x50	2.94 74	1.94 49	11.13 283	11.75 298	11.13 283	3.6 91	5.3 135	8.3 210	3.3 83	4.6 117	8.3 210	70 32	51 23
4x3 100x80	3.94 100	2.94 74	12 305	12.63 321	12 305	5 128	6.3 161	10 255	5 128	5.6 141	10 255	119 54	90 41
6x4 150x100	5.94 150	3.94 100	15.88 403	16.5 419	18 457	6.3 160	7.5 190	12.6 320	6.3 160	6.7 170	12.6 320	220 100	165 75
8x6 200x150	7.94 201	5.94 150	19.75 518	20.38 518	20.5 521	7.5 190	9.2 233	15 380	7.5 190	8.7 222	15 380	495 225	451 205
10x8 250x200	9.94 252	7.94 201	22.38 568	23 584	22 559	9 228	10 253	17.5 445	9.3 235	10 255	17.5 445	803 365	748 340
12x10 300x250	11.94 303	9.94 252	25.5 648	26.13 664	25 635	10.6 268	11.4 290	20.5 520	11.4 290	12 305	20.5 520	1320 600	1239 563
14x10 350x250	13.19 334	9.94 252	30 762	30.63 778	30 762	10.6 268	11.4 290	23 585	11.4 290	12 305	23 585	1430 650	1327 603
16x12 400x300	15.19 385	11.94 334	33 838	33.63 854	33 838	12.9 327	15.6 395	23.2 590	13.4 340	13.4 340	21.8 554	1782 810	1586 721
18x14 450x350	17.19 436	13.19 334	36 914	36.62 930	36 914	15.2 385	15.7 400	28 710	14.5 368	15.1 384	28 710	3300 1500	3102 1410
20x16 500x400	19.19 487	15.19 385	39 991	49.75 1010	39 991	16 405	17.8 452	28.2 716	16.5 420	16.5 419	27.1 689	4290 1950	3630 1650
24x20 600x500	23.19 589	19.19 487	45 1143	45.88 1165	45 1143	21.7 550	21.7 550	36 915	19.5 495	19.9 505	36 915	5500 2500	4950 2250
30x24 750x600	28.94 735	23.19 589	55 1397	56 1422	55 1397	24 610	24.8 630	42.9 1090	24.85 630	23.9 607	42.9 1090	9350 4250	8492 3860
36x30 900x750	34.44 874	28.94 735	55 1397	56 1422	55 1397	28 710	30.6 777	50 1270	29.4 747	29.4 747	50 1270	13900 6350	12892 5860

(1) Contact our sales representative for more information.

TRUNNION BALL VALVE

DIMENSIONS AND WEIGHTS

ASME CLASS 600 (PN 100)



ASME CLASS 600 (PN 100) FULL BORE

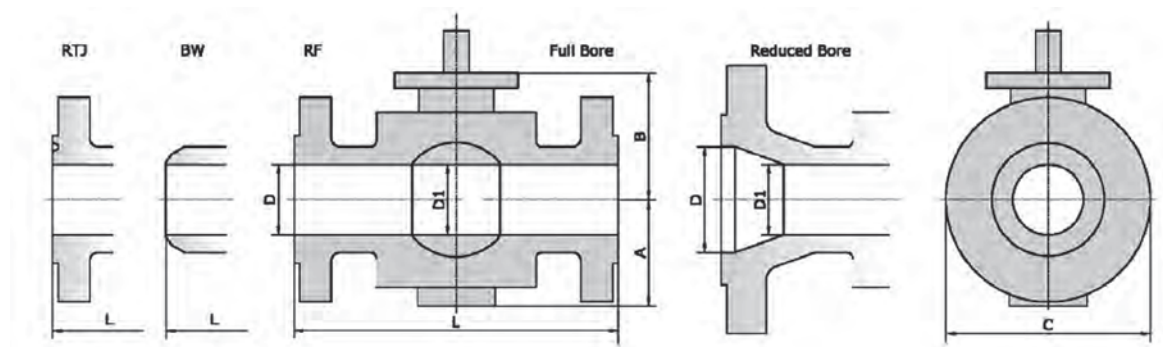
Size in/mm	D	D1	L			Casting			Forging			Weight	
			RF	RTJ	BW	A	B	C	A	B	C	RF/RTJ (lb/kg)	BW (lb/kg)
2	1.93	1.93	11.5	11.63	11.5	3.7	5.7	6.5	3.3	5	6.5	73	62
50	49	49	292	295	292	95	145	165	83	126	165	33	28
2 1/2	2.44	2.44	13	13.13	13	3.9	6.3	7.7	3.7	5.9	7.4	119	106
65	62	62	330	333	330	100	160	195	95	150	188	54	48
3	2.94	2.94	14	14.13	14	5	7.2	8.3	4.6	6.1	8.3	150	110
80	74	74	356	359	356	128	184	210	117	155	210	68	50
4	3.94	3.94	17	17.13	17	5.8	8.2	11.6	5.4	7	10.8	242	209
100	100	100	432	435	432	148	209	295	138	177	275	110	95
5	5	5	20	20.12	20	6.7	8.2	13.4	6.5	8	13	407	352
125	127	125	508	511	508	170	208	340	165	202	330	185	160
6	5.94	5.94	22	22.13	22	7.7	9.8	14	7	9	14	528	444
150	150	150	559	562	559	195	250	355	178	228	355	240	202
8	7.91	7.91	26	26.13	26	9.6	11.1	16.5	9.6	10.5	16.5	796	638
200	201	201	660	663	660	245	283	420	245	267	420	362	290
10	9.92	9.92	31	31.13	31	11.4	12.6	20	11.9	12.3	20	1166	898
250	252	252	787	790	787	290	320	510	302	312	510	530	408
12	11.94	11.94	33	33.13	33	14.2	15.3	22.8	13.8	14.5	22.6	1727	1430
300	303	303	838	841	835	360	390	580	350	368	573	785	650
14	13.19	13.19	35	35.12	35	15.6	16.1	25	15	15.6	24.8	2376	2002
350	334	334	889	891	889	395	410	635	380	396	630	1080	910
16	15.19	15.19	39	39.13	39	16.7	17.2	28.3	17.3	17.5	16.7	3564	3080
400	385	385	991	994	991	425	436	720	440	444	423	1620	1400
18	17.19	17.19	43	43.13	43	19.3	20.3	32.3	18.9	20	31.9	3982	3630
450	436	435	1092	1095	1092	490	515	820	480	510	810	1810	1650
20	19.17	19.17	47	47.25	47	22.6	22.7	36.2	20.6	20.7	34.4	5500	4818
500	487	487	1194	1200	1194	575	576	920	522	525	875	2500	2190
22	21.19	21.19	51	51.38	51	20.5	22	37.8	20.3	26.7	37.4	6270	5544
550	538	538	1295	1305	1295	520	560	960	515	550	950	2850	2520
24	23.19	23.19	55	55.38	55	23.8	25.2	41	24.4	25.2	41.3	8492	7612
600	589	589	1397	1407	1397	605	640	1043	619	641	1050	3860	3460
26	24.94	24.94	57	57.5	57	25.6	26.4	43.9	25.5	26.2	43.3	9504	8470
650	633	633	1448	1461	1448	650	670	1115	640	665	1100	4320	3850
28	26.94	26.94	61	61.5	61	26.8	23.3	48	26.6	28.1	47.6	12848	11660
700	684	684	1549	1562	1594	680	720	1220	675	715	1210	5840	5300
30	28.94	28.94	65	65.5	65	29.9	30.6	51.8	29.1	30.6	51	15686	14080
750	735	735	1651	1664	1651	760	777	1315	740	776	1295	7130	6400
32	30.69	30.69	70	70	70.63	31.1	31.5	55.5	30.9	31.3	53.9	20240	18260
800	779	779	1778	1778	1794	790	790	1410	785	794	1370	9200	8300
34	32.69	32.69	76	76.61	76	33.1	33.9	56.3	32.7	33.5	57.5	22660	20900
850	830	830	1930	1946	1930	840	860	1480	830	850	1460	10300	9500
36	34.41	34.41	82	82.63	82	35	36.6	60.9	35.4	36.2	60.9	25740	23100
900	874	874	2083	2099	2083	890	930	1546	900	920	1548	11700	10500
40	38.44	38.44	80.71	(1)	80.71	37.8	38	65.6	37.4	37.6	65.2	31460	29172
1000	976	976	2050	(1)	2050	960	965	1665	950	955	1655	14300	13260
42	40.19	40.19	82.68	(1)	82.68	40.9	41.7	72.8	40.6	41.3	71.7	36080	33660
1050	1020	1020	2100	(1)	2100	1040	1060	1850	1030	1050	1820	16400	15300
48	45.94	45.94	94.49	(1)	94.49	46.9	46.3	78.3	46.5	45.9	77.6	49940	46200
1200	1166	1166	2400	(1)	2400	1190	1175	1990	1180	1165	1970	22700	21000
56	53.56	53.56	106.3	(1)	106.3	54.5	54.3	93.5	54.2	53.5	92.6	81400	30800
1400	1360	1360	2700	(1)	2700	1385	1380	2375	1376	1360	2352	37000	14000

(1) Contact our sales representative for more information.

TRUNNION BALL VALVE

DIMENSIONS AND WEIGHTS

ASME CLASS 600 (PN 100)



ASME CLASS 600 (PN 100) REDUCED BORE

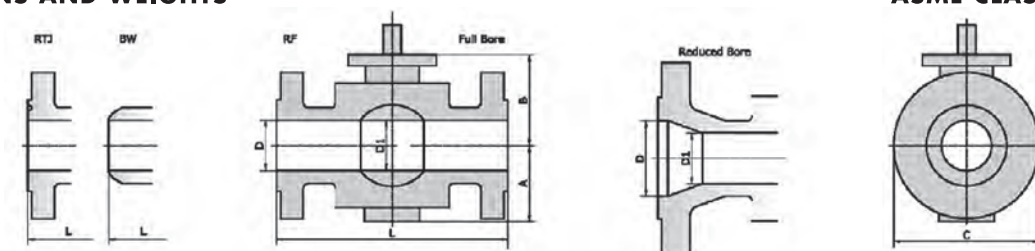
Size in/mm	D	D1	L			Casting			Forging			Weight	
			RF	RTJ	BVV	A	B	C	A	B	C	RF/RTJ (lb/kg)	BVV (lb/kg)
2x1 1/2 50x40	1.94 49	1.5 38	11.5 292	11.62 295	11.5 292	3.7 94	5.28 134	6.5 165	3.3 83	4.5 115	6.5 165	48 22	35 16
2 1/2x2 65x50	2.44 62	1.94 49	13 330	13.13 333	13 330	3.7 95	5.7 145	7.5 190	3.3 83	5 126	7.5 190	64 29	55 25
3x2 80x50	2.94 74	1.94 49	14 356	14.13 359	14 356	3.7 95	5.7 145	8.3 210	3.3 83	5 126	8.3 210	81 37	55 25
4x3 100x80	3.94 100	2.94 74	17 432	17.13 435	17 432	5.4 138	7.2 184	10.8 275	5.5 138	6.1 155	10.8 275	163 74	110 50
6x4 150x100	5.94 150	3.94 100	22 559	22.13 562	22 559	7 178	8.2 208	14 355	7 178	7 177	14 355	330 150	187 85
8x6 200x150	7.94 201	5.94 150	26 660	26.13 663	26 660	8.3 210	9.8 250	16.5 420	8.3 210	9 228	16.5 420	638 290	528 240
10x8 250x200	9.94 252	7.94 201	31 787	31.13 790	31 787	9.6 245	11.1 283	20 510	9.6 245	10.5 267	20 510	1067 485	880 400
12x10 300x250	11.94 303	9.94 252	33 838	33.13 841	33 838	11.4 290	12.6 320	22 560	11.9 302	12.3 312	22 560	1463 665	1100 500
14x10 350x250	13.19 334	9.94 252	35 889	35.13 892	35 889	11.4 290	12.6 320	23.8 605	11.9 302	12.3 312	23.8 605	2266 1030	2002 910
16x12 400x300	15.19 385	11.94 303	39 991	39.13 994	39 991	14.1 360	15.4 390	27 685	13.8 350	14.5 368	27 685	2530 1150	2200 1000
18x14 450x350	17.19 436	13.19 334	43 1092	43.13 1095	43 1092	15.6 395	16.1 410	29.3 745	15.2 385	15.6 396	29.3 745	3630 1650	2860 1300
20x16 500x400	19.19 487	15.19 385	47 1194	47.25 1200	47 1194	16.7 425	17.2 436	32 815	17.3 440	17.5 444	32.1 815	5632 2560	4620 2100
24x20 600x500	23.19 589	19.19 487	55 1397	55.38 1407	55 1397	22.6 575	22.7 576	37 940	20.6 522	20.7 525	37 940	6710 3050	5720 2600
30x24 750x600	28.94 735	23.19 589	61 1651	65.5 1664	65 1651	23.8 605	25.2 640	42.9 1090	24.4 619	25.2 641	42.9 1090	10780 4900	9240 4200
36x30 900x750	34.41 874	28.94 735	82 2083	82.63 2099	82 2083	35 890	36.6 930	60.9 1546	35.4 900	36.2 920	60.9 1548	20020 9100	17600 8000

(1) Contact our sales representative for more information.

TRUNNION BALL VALVE

DIMENSIONS AND WEIGHTS

ASME CLASS 900 (PN 150)



ASME CLASS 900 (PN 150) FULL BORE

Size in/mm	D	D1	L			Casting			Forging			Weight	
			RF	RTJ	BW	A	B	C	A	B	C	RF/RTJ (lb/kg)	BW (lb/kg)
2	1.94	1.94	14.5	14.63	14.5	4.3	5.5	8.5	4.3	5.5	8.5	128	75
50	49	49	368	371	368	108	140	215	108	140	215	58	34
2 1/2	2.44	2.44	16.5	16.62	16.5	4.8	6.3	9.6	4.8	6.3	9.6	165	123
65	62	62	419	422	419	123	160	245	123	160	245	75	56
3	2.94	2.94	15	15.13	15	4.7	6.3	9.4	4.7	6.3	9.4	183	152
80	74	74	381	384	381	120	161	240	120	161	240	83	69
4	3.94	3.94	18	18.13	18	5.7	7.1	11.4	5.7	7.1	11.4	356	319
100	100	100	457	460	457	145	180	290	145	180	290	162	145
5	4.76	4.76	22	22.13	22	6.7	8.5	13.8	6.7	8.5	13.8	517	429
125	121	121	559	562	559	170	215	350	170	215	350	235	195
6	5.94	5.94	24	24.13	24	7.3	9.4	15	7.3	9.4	15	715	517
150	150	150	610	613	610	185	239	380	185	239	380	325	235
8	7.94	7.94	29	29.13	29	7.7	11.3	18.5	7.7	11.3	18.5	1047	772
200	201	201	737	740	737	295	288	470	295	288	470	476	351
10	9.94	9.94	33	33.13	33	13.8	13.8	21.5	13.8	13.8	21.5	1969	1650
250	252	252	838	841	838	350	350	545	350	350	545	895	750
12	11.94	11.94	38	38.13	38	14.6	15.3	24	14.6	15.3	24	2519	2189
300	303	303	965	968	965	370	388	610	370	388	610	1145	995
14	12.69	12.69	40.5	40.88	40.5	16.9	16.9	26.2	16.1	16.1	25.9	3278	2519
350	322	322	1029	1039	1029	430	430	665	410	410	658	1490	1145
16	14.69	14.69	44.5	44.88	44.5	17.7	18.1	29.5	17.7	18.1	29.5	4092	3190
400	373	373	1130	1140	1130	450	460	750	450	460	750	1860	1450
18	16.69	16.69	48	48.5	48	19.3	20.9	32.3	19.3	20.9	32.3	5841	4554
450	423	423	1219	1232	1219	490	530	820	490	530	820	2655	2070
20	18.56	18.56	52	52.5	52	23.2	22.2	36.5	22.2	22.2	36.5	6952	5357
500	471	471	1321	1334	1321	590	565	928	590	565	928	3160	2435
24	22.44	22.44	61	61.75	61	24.8	24.2	41.7	24.8	24.2	41.7	11220	8360
600	570	570	1549	1568	1549	630	615	1060	630	615	1060	5100	3800
26	24.31	24.31	65	65.88	(1)	26.8	27.6	45.7	26.8	27.6	45.7	14850	11968
650	617	617	1651	1673	(1)	680	700	1160	680	700	1160	6750	5440
28	26.19	26.19	69	69.88	(1)	29.3	29.5	50.6	29.3	29.5	50.6	19250	16390
700	665	665	1753	1775	(1)	743	750	1285	743	750	1285	8750	7450
30	28.06	28.06	74	74.88	(1)	31.5	31.5	53.5	31.5	31.5	53.5	25300	21340
750	712	712	1880	1902	(1)	800	800	1360	800	800	1360	11500	9700

ASME CLASS 900 (PN 150) REDUCED BORE

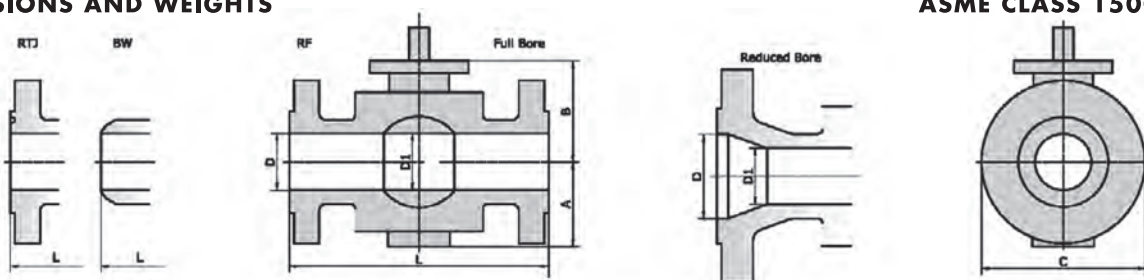
Size in/mm	D	D1	L			Casting			Forging			Weight	
			RF	RTJ	BW	A	B	C	A	B	C	RF/RTJ (lb/kg)	BW (lb/kg)
2x1 1/2	1.94	1.5	14.5	14.62	14.5	4.3	4.5	8.5	4.3	4.5	8.5	99	55
50x40	49	38	215	371	368	108	115	215	108	115	215	45	25
2 1/2x2	2.44	1.94	16.5	16.63	16.5	4.3	5.1	9.6	4.3	5.1	9.6	123	81
65x50	62	49	419	422	419	108	140	245	108	140	245	56	37
3x2	2.94	1.94	15	15.13	15	4.3	5.1	9.4	4.3	5.1	9.4	143	110
80x50	74	49	381	384	381	108	140	240	108	140	240	65	50
4x3	3.94	2.94	18	18.13	18	5.7	6.3	11.4	5.7	6.3	11.4	238	172
100x80	100	74	457	460	457	145	161	290	145	161	290	108	78
6x4	5.94	3.94	24	24.13	24	7.5	7.1	15	7.5	7.1	15	462	330
150x100	150	100	610	613	610	190	180	380	190	180	380	210	150
8x6	7.94	5.94	29	29.13	29	9.3	9.4	18.5	9.3	9.4	18.5	715	528
200x150	201	150	737	740	737	235	239	470	235	239	470	325	240
10x8	9.94	7.94	33	33.13	33	11.3	11.3	21.5	11.3	11.3	21.5	902	638
250x200	252	201	838	841	838	295	288	545	295	288	545	410	290
12x10	11.94	9.94	38	38.13	38	13	13.8	24	13	13.8	24	2420	1936
300x250	303	252	965	968	965	330	350	610	330	350	610	1100	880
14x10	14.69	11.94	44.5	44.88	44.5	14.6	15.3	24	14.6	15.3	24	3190	2816
350x250	322	303	1130	1140	1130	370	388	610	370	388	610	1450	1280
16x12	14.69	11.94	44.5	44.88	44.5	14.6	15.6	24	14.6	15.6	24	3608	2970
400x300	373	303	1130	1140	1130	370	388	610	370	388	610	1640	1350
18x14	16.69	12.69	48	48.5	48	16.9	16.9	30.9	16.1	16.1	30.9	5280	4290
450x350	423	322	1219	1232	1219	430	430	785	410	410	785	2400	1950
20x16	18.56	14.69	52	52.5	52	17.7	18.1	33.7	17.7	18.1	33.7	6292	4730
500x400	471	373	1321	1334	1321	450	460	855	450	460	855	2860	2150
24x20	22.44	18.56	61	61.75	61	23.2	22.2	40.9	23.2	22.2	40.9	9240	6336
600x500	570	471	1549	1568	1549	590	565	1040	590	565	1040	4200	2880
30x24	28.06	22.44	74	74.88	(1)	24.8	24.2	48.4	24.8	24.2	48.4	(1)	(1)
750x600	712	570	1880	1902	(1)	630	615	1230	630	615	1230	(1)	(1)

(1) Contact our sales representative for more information.

TRUNNION BALL VALVE

DIMENSIONS AND WEIGHTS

ASME CLASS 1500 (PN 250)



ASME CLASS 1500 (PN 250) FULL BORE

Size in/mm	D	D1	L			Casting			Forging			Weight	
			RF	RTJ	BW	A	B	C	A	B	C	RF/RTJ (lb/kg)	BW (lb/kg)
2	1.94	1.94	14.5	14.63	14.5	4.3	5.5	8.5	4.3	5.5	8.5	128	75
50	49	49	368	371	368	108	140	215	108	140	215	58	34
2 1/2	2.44	2.44	16.5	16.62	16.5	4.8	6.3	9.6	4.8	6.3	9.6	174	136
65	62	62	419	422	419	123	160	245	123	160	245	79	62
3	2.94	2.94	18.5	18.63	18.5	5.5	7.4	10.4	5.5	7.4	10.4	216	154
80	74	74	470	473	470	140	187	265	140	187	265	98	70
4	3.94	3.94	21.5	21.63	21.5	6.5	7.9	12.2	6.5	7.9	12.2	365	317
100	100	100	546	549	546	165	201	310	165	201	310	166	144
5	4.37	4.37	26.5	27.04	26.5	6.9	8.7	14.8	6.9	8.7	14.8	847	642
125	111	111	673	676	673	175	220	375	175	220	375	385	292
6	5.69	5.69	27.75	28	27.75	9.6	11.3	16.3	9.6	11.3	16.3	1030	834
150	144	144	705	711	705	219	288	415	219	288	415	468	379
8	7.56	7.56	32.75	33.13	32.75	11.6	12.8	19.1	11.6	12.8	19.1	1478	964
200	192	191	832	842	832	295	325	485	295	325	485	672	438
10	9.44	9.44	39	39.38	39	14.2	15.9	23.6	14.2	15.9	23.6	2222	1716
250	239	239	991	1001	991	360	405	600	360	405	600	1010	780
12	11.31	11.31	44.5	45.13	44.5	15.7	17	27.2	15.7	17	27.2	3388	2420
300	287	287	1130	1146	1130	400	432	692	400	432	692	1540	1100
14	12.44	12.44	49.5	50.25	49.5	19.3	18.9	30.7	18.7	18.5	30.1	6820	5280
350	315	315	1259	1276	1257	490	480	780	475	469	765	3100	2400
16	14.19	14.19	54.5	55.38	54.5	24.8	23.5	38.6	24.8	23.5	38.6	9900	6072
400	360	360	1384	1407	1384	629	598	980	629	598	980	4500	2760
18	16	16	60.5	61.38		24	24.4	39.4	24	24.4	39.4	13420	10670
450	406	406	1537	1559	(1)	610	620	1000	610	620	1000	6100	4850
20	17.88	17.88	65.5	66.38		25.6	25.6	41.9	25.6	25.6	41.9	16720	12540
500	454	454	1664	1686	(1)	650	650	1065	650	650	1065	7600	5700
24	21.5	21.5	76.5	77.63		29.9	31.4	50.2	29.9	31.4	50.2	21120	15840
600	546	546	1943	1972	(1)	760	798	1275	760	798	1275	9600	7200

ASME CLASS 1500 (PN 250) REDUCED BORE

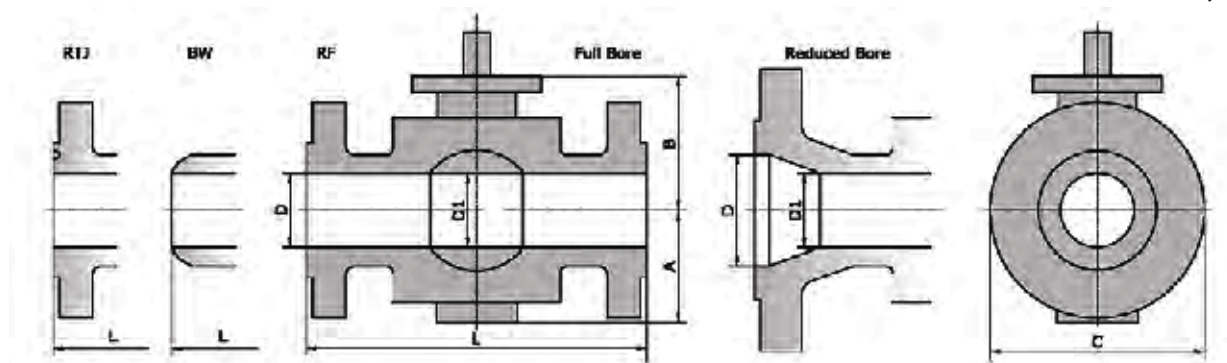
Size in/mm	D	D1	L			Casting			Forging			Weight	
			RF	RTJ	BW	A	B	C	A	B	C	RF/RTJ (lb/kg)	BW (lb/kg)
2x1 1/2	1.94	1.5	14.5	14.62	14.5	4.3	4.5	8.5	4.3	4.5	8.5	95	48
50x40	49	38	368	371	368	108	115	215	108	115	215	43	22
2 1/2x2	2.44	1.94	16.5	16.63	16.5	4.3	5.5	9.6	4.3	5.5	9.6	141	92
65x50	62	49	419	422	419	108	140	245	108	140	245	64	42
3x2	2.94	1.94	18.5	18.63	18.5	4.3	5.5	10.6	4.3	5.5	10.6	165	106
80x50	74	49	470	473	470	108	140	265	108	140	265	75	48
4x3	3.94	2.94	21.5	21.63	21.5	6.1	7.4	12.2	6.1	7.4	12.2	286	167
100x80	100	74	546	549	546	155	187	310	155	187	310	130	76
6x4	5.69	3.94	27.75	28	27.75	7.8	7.9	15.6	7.8	7.9	15.6	605	374
150x100	144	100	705	711	705	198	201	395	198	201	395	275	170
8x6	7.56	5.69	32.75	33.13	32.75	9.6	11.3	19.1	9.6	11.3	19.1	1320	924
200x150	192	144	832	841	832	243	288	485	243	288	485	600	420
10x8	9.44	7.56	39	39.38	39	11.6	12.8	23	11.6	12.8	23	1848	1078
250x200	239	192	991	1001	991	295	325	585	295	325	585	840	490
12x10	11.31	9.44	44.5	45.13	44.5	14.2	15.9	26.6	14.2	15.9	26.6	2750	1760
300x250	287	239	1130	1146	1130	360	405	675	360	405	675	1250	800
14x10	12.44	9.44	49.5	50.25	49.5	14.2	15.9	29.5	14.2	15.9	29.5	3630	2970
350x250	315	239	1257	1276	1257	360	405	750	360	405	750	1650	1350
16x12	14.19	11.31	54.5	55.38	54.5	15.7	17	32.5	15.7	17	32.5	4517	3520
400x300	360	287	1384	1407	1384	400	432	825	400	432	825	2053	1600
18x14	16	12.44	60.5	61.38	60.5	18.7	18.5	36	18.7	18.5	36	10956	8690
450x350	406	315	1537	1559	1537	475	469	915	475	469	915	4980	3950
20x16	17.88	14.19	65.5	66.38		24.8	23.5	38.8	24.8	23.5	38.8	12188	9350
500x400	454	360	1664	1686	(1)	629	598	985	629	598	985	5540	4250
24x20	21.5	17.88	76.5	77.63	76.5	25.6	25.6	46.1	25.6	25.6	46.1	24200	18920
600x500	546	454	1943	1971	1943	650	650	1170	650	650	1170	11000	8600

(1) Contact our sales representative for more information.

TRUNNION BALL VALVE

DIMENSIONS AND WEIGHTS

ASME CLASS 2500 (PN 420)



ASME CLASS 2500 (PN 420) FULL BORE

Size in/mm	D	D1	L			Casting			Forging			Weight	
			RF	RTJ	BW	A	B	C	A	B	C	RF/RTJ (lb/kg)	BW (lb/kg)
2	1.69	1.69	17.75	17.88	17.75	/	/	/	5.5	7.2	9.3	260	154
50	42	42	451	454	451	/	/	/	139	183	235	118	70
2 1/2	2.06	2.06	20.32	20.56	20.32	/	/	/	5.2	6.8	10.4	378	275
65	52	52	508	514	508	/	/	/	133	173	265	172	125
3	2.44	2.44	22.75	23	22.75	/	/	/	6.3	8.1	12	484	363
80	62	62	578	584	578	/	/	/	160	205	305	220	165
4	3.44	3.44	26.5	26.88	26.5	/	/	/	7.4	6.8	14	803	715
100	87	87	673	683	673	/	/	/	188	173	355	365	325
5	3.62	3.62	31.26	31.77	31.26	/	/	/	7.1	10	16.5	1342	1078
125	92	92	794	807	794	/	/	/	180	255	420	610	490
6	5.19	5.19	36	36.5	36	/	/	/	10.8	13.4	19.1	1650	1375
150	131	131	914	927	914	/	/	/	275	341	485	750	625
8	7.06	7.06	40.25	40.88	40.25	/	/	/	15	16.5	24.2	4620	3608
200	179	179	1022	1038	1022	/	/	/	380	418	615	2100	1640
10	8.81	8.81	50	50.88	50	/	/	/	18.9	20.5	31.9	6600	5588
250	223	223	1270	1292	1270	/	/	/	480	520	810	3000	2540
12	10.44	10.44	56	56.88	56	/	/	/	19.7	21.3	33.9	9240	7700
300	265	265	1422	1444	1422	/	/	/	500	540	860	4200	3500

ASME CLASS 2500 (PN 420) REDUCED BORE

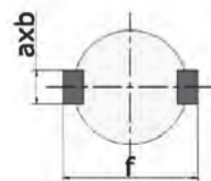
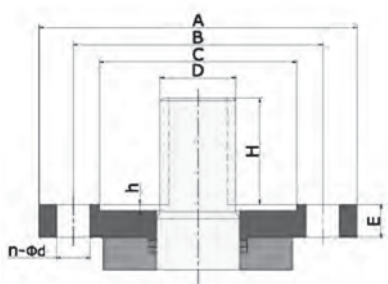
Size in/mm	D	D1	L			Casting			Forging			Weight	
			RF	RTJ	BW	A	B	C	A	B	C	RF/RTJ (lb/kg)	BW (lb/kg)
2x1 1/2	1.69	1.5	17.75	17.88	17.75	/	/	/	4.6	5.5	9.3	189	97
50x40	42	38	368	454	451	/	/	/	118	140	235	86	44
2 1/2x2	2.06	1.69	20	20.25	20	/	/	/	5.5	7.2	10.4	297	242
65x50	52	42	508	514	508	/	/	/	139	183	265	135	110
3x2	2.44	1.69	22.75	23	22.75	/	/	/	5.5	7.2	12	341	220
80x50	62	42	578	584	578	/	/	/	139	183	305	155	100
4x3	3.44	2.66	26.5	26.88	26.5	/	/	/	7	10.4	14	627	374
100x80	87	62	673	683	673	/	/	/	178	265	355	285	170
6x4	5.19	3.44	36	36.5	36	/	/	/	9.6	10.7	19.1	1254	737
150x100	131	87	914	927	914	/	/	/	243	273	485	570	335
8x6	7.06	5.19	40.25	40.88	40.25	/	/	/	10.8	13.4	21.7	2332	1914
200x150	179	131	1022	1038	1022	/	/	/	275	341	550	1060	870
10x8	8.81	7.06	50	50.88	50	/	/	/	15	16.5	26.6	3740	2882
250x200	223	179	1270	1292	1270	/	/	/	380	418	675	1700	1310
12x10	10.44	8.81	56	56.88	56	/	/	/	18.9	20.5	31.9	5654	4510
300x250	265	223	1422	1445	1422	/	/	/	480	520	810	2570	2050

(1) Contact our sales representative for more information.

TRUNNION BALL VALVE

TOP MOUNTING DIMENSIONS AND TORQUE

NPS 2-4 (DN 50-100)



Size in/mm	Rating Class/PN	Torque ⁽¹⁾ ft·lb/N·m	ISO F#	A	B	C	D	h	E	n-Φd ⁽²⁾	H	Drive Key a x b	f	Gearbox Model
2 50	150	44	F 10	4.92	4.02	2.76	0.945	0.118	0.63	4-Φ0.472	1.575	0.315 x 0.276	1.063	GE-A1
	20	60		125	102	70	24	3	16	4-Φ 12	40	8 x 7	27	
	300	81	F 10	4.92	4.02	2.76	0.945	0.118	0.63	4-Φ0.472	1.575	0.315 x 0.276	1.063	GE-A1
	50	110		125	102	70	24	3	16	4-Φ 12	40	8 x 7	27	
	600	111	F 10	4.92	4.02	2.76	0.945	0.118	0.63	4-Φ0.472	1.575	0.315 x 0.276	1.063	GE-A1
	100	150		125	102	70	24	3	16	4-Φ 12	40	8 x 7	27	
	900	133	F 12	5.91	4.92	3.35	1.102	0.118	0.83	4-Φ0.551	1.772	0.315 x 0.276	1.063	GE-A2
	150	180		150	125	85	28	3	21	4-Φ 14	45	8 x 7	27	
	1500	148	F 12	5.91	4.92	3.35	1.102	0.118	0.83	4-Φ0.551	1.772	0.315 x 0.276	1.063	GE-A2
	250	200		150	125	85	28	3	21	4-Φ 14	45	8 x 7	27	
2 ½ 65	2500	295	F 16	8.27	6.5	5.12	1.417	0.118	1.10	4-Φ0.866	2.362	0.394 x 0.315	1.339	GE-A4
	420	400		210	165	130	36	3	28	4-Φ 22	60	10 x 8	34	
	150	59	F 10	4.92	4.02	2.76	0.945	30.118	0.63	4-Φ0.472	1.575	0.315 x 0.276	1.063	GE-A1
	20	80		125	102	70	24	3	16	4-Φ 12	40	8 x 7	27	
	300	96	F 10	4.92	4.02	2.76	0.945	30.118	0.63	4-Φ0.472	1.575	0.315 x 0.276	1.063	GE-A1
	50	130		125	102	70	24	3	16	4-Φ 12	40	8 x 7	27	
	600	133	F 12	5.91	4.92	3.35	1.102	0.118	0.83	4-Φ0.551	1.772	0.315 x 0.276	1.063	GE-A2
	100	180		150	125	85	28	3	21	4-Φ 14	45	8 x 7	27	
	900	177	F 12	5.91	4.92	3.35	1.102	0.118	0.83	4-Φ0.551	1.772	0.315 x 0.276	1.063	GE-A2
	150	240		150	125	85	28	3	21	4-Φ 14	45	8 x 7	27	
3 80	1500	221	F 14	6.89	5.51	3.94	1.417	0.157	0.98	4-Φ0.709	2.362	0.394 x 0.315	1.339	GE-A3
	250	300		175	140	100	36	4	25	4-Φ 18	60	10 x 8	34	
	2500	354	F 16	8.27	6.5	5.12	1.575	0.197	1.18	4-Φ0.866	2.362	0.472 x 0.315	1.693	GE-A4
	420	480		210	165	130	40	5	30	4-Φ 22	60	12 x 8	43	
	150	74	F 10	4.92	4.02	2.76	0.945	0.118	0.87	4-Φ0.472	1.575	0.315 x 0.276	1.063	GE-A1
	20	100		125	102	70	24	3	22	4-Φ 12	40	8 x 7	27	
	300	96	F 10	4.92	4.02	2.76	0.945	0.118	0.87	4-Φ0.472	1.575	0.315 x 0.276	1.063	GE-A1
	50	130		125	102	70	24	3	22	4-Φ 12	40	8 x 7	27	
	600	148	F 12	5.91	4.92	3.35	1.102	0.118	0.91	4-Φ0.551	1.772	0.315 x 0.276	1.063	GE-A2
	100	200		150	125	85	28	3	23	4-Φ 14	45	8 x 7	27	
4 100	900	207	F 14	6.89	5.51	3.94	1.417	0.157	0.98	4-Φ0.709	2.362	0.394 x 0.315	1.339	GE-A3
	150	280		175	140	100	36	4	25	4-Φ 18	60	10 x 8	34	
	1500	354	F 14	6.89	5.51	3.94	1.575	0.157	1.14	4-Φ0.709	2.362	0.472 x 0.315	1.693	GE-A3
	250	480		175	140	100	40	4	29	4-Φ 18	60	12 x 8	43	
	2500	443	F 16	8.27	6.5	5.12	1.772	0.197	1.18	4-Φ0.866	2.756	0.551 x 0.354	1.949	GE-A4
	420	600		210	165	130	45	5	30	4-Φ 22	70	14 x 9	49.5	
	150	89	F 12	5.91	4.92	3.35	1.102	0.118	0.91	4-Φ0.551	1.772	0.315 x 0.276	1.063	GE-A2
	20	120		150	125	85	28	3	23	4-Φ 14	45	8 x 7	27	
	300	133	F 12	5.91	4.92	3.35	1.102	0.118	0.91	4-Φ0.551	1.772	0.315 x 0.276	1.063	GE-A2
	50	180		150	125	85	28	3	23	4-Φ 14	45	8 x 7	27	
4 100	600	207	F 14	6.89	5.51	3.94	1.417	0.157	0.98	4-Φ0.709	2.362	0.394 x 0.315	1.339	GE-A3
	100	280		175	140	100	36	4	25	4-Φ 18	60	10 x 8	34	
	900	369	F 14	6.89	5.51	3.94	1.417	0.157	0.98	4-Φ0.709	2.362	0.394 x 0.315	1.339	GE-A3
	150	500		175	140	100	36	4	25	4-Φ 18	60	10 x 8	34	
	1500	590	F 16	8.27	6.5	5.12	1.772	0.197	1.18	4-Φ0.866	2.756	0.551 x 0.354	1.949	GE-A4
	250	800		210	165	130	45	5	30	4-Φ 22	70	14 x 9	49.5	
	2500	1475	F 16	8.27	6.5	5.12	1.969	0.197	1.18	4-Φ0.866	2.756	0.551 x 0.354	2.244	GE-A5
	420	2000		210	165	130	50	5	30	4-Φ 22	70	14 x 9	57	

Note:

(1) Torque doesn't include safety factor. The torque is based on seats of PTFE/RPTFE (Class 150 - 300), Nylon (Class 900 - 1500), and PEEK (Class 2500).

Torque shall be as follows if other than the base seat material is selected:

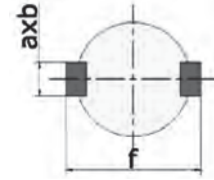
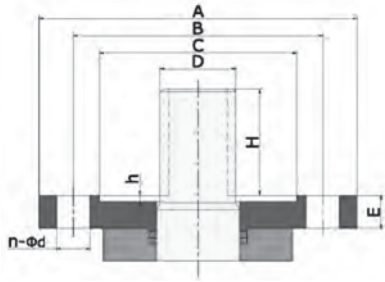
PTFE / RPTFE	Nylon	Devlon	PCTFE	TFM	Delrin	Molon	PPL	PEEK	Metal
Base	x 1	x 1	x 0.8	X 0.8	x 1	X 1	X 0.8	x 1.3	x 2.2

(2) Number of bolt holes and bolt hole diameter.

TRUNNION BALL VALVE

TOP MOUNTING DIMENSIONS AND TORQUE (CONT'D)

NPS 6-10 (DN 150-300)



Size in/mm	Rating Class/PN	Torque ⁽¹⁾ ft·lb/N·m	ISO F#	A	B	C	D	h	E	n-Φd ⁽²⁾	H	Drive Key a x b	f	Gearbox Model
6 150	150	369	F14	6.89	5.51	3.94	1.417	0.157	0.984	4-Φ0.709	2.362	0.551 x 0.315	1.339	GE-A3
	20	500		175	140	100	36	4	25	4-Φ 18	60	10 x 8	34	
	300	516	F14	6.89	5.51	3.94	1.417	0.157	0.984	4-Φ0.709	2.362	0.551 x 0.315	1.339	GE-A3
	50	700		175	140	100	36	4	25	4-Φ 18	60	10 x 8	34	
	600	811	F16	8.27	6.5	5.12	1.772	0.197	1.181	4-Φ0.866	2.756	0.551 x 0.354	1.949	GE-A4
	100	1100		210	165	130	45	5	30	4-Φ 22	70	14 x 9	49.5	
	900	1549	F16	8.27	6.5	5.12	1.969	0.197	1.181	4-Φ0.866	2.756	0.551 x 0.354	2.244	GE-A5
	150	2100		210	165	130	50	5	30	4-Φ 22	70	14 x 9	57	
	1500	2213	F16	8.27	6.5	5.12	2.165	0.197	1.181	4-Φ0.866	3.15	0.63 x 0.394	2.48	GE-A6
	250	3000		210	165	130	55	5	30	4-Φ 22	80	16 x 10	63	
8 200	2500	3688	F25	11.81	10	7.87	2.559	0.197	1.378	4-Φ0.709	3.543	0.709 x 0.433	2.874	GE-A9
	420	5000		300	254	200	65	5	35	4-Φ 18	90	18 x 11	73	
	150	443	F16	8.27	6.5	5.12	1.772	0.197	1.181	4-Φ0.866	2.756	0.551 x 0.354	1.949	GE-A4
	20	600		210	165	130	45	5	30	4-Φ 22	70	14 x 9	49.5	
	300	811	F16	8.27	6.5	5.12	1.772	0.197	1.181	4-Φ0.866	2.756	0.551 x 0.354	1.949	GE-A4
	50	1100		210	165	130	45	5	30	4-Φ 22	70	14 x 9	49.5	
	600	1401	F16	8.27	6.5	5.12	1.969	0.197	1.181	4-Φ0.866	2.756	0.551 x 0.354	2.244	GE-A6
	100	1900		210	165	130	50	5	30	4-Φ 22	70	14 x 9	57	
	900	2581	F25	11.81	10	7.87	2.165	0.197	1.181	8-Φ0.709	3.15	0.63 x 0.394	2.48	GE-A8
	150	3500		300	254	200	55	5	30	8-Φ 18	80	16 x 10	63	
10 250	1500	4794	F25	11.81	10	7.87	2.756	0.197	1.496	8-Φ0.709	3.937	0.787 x 0.472	3.11	GE-B1
	250	6500		300	254	200	70	5	38	8-Φ 18	100	20 x 12	79	
	2500	6269	F30	13.78	11.73	9.06	3.543	0.197	1.496	8-Φ0.866	4.724	0.984 x 0.551	3.937	GE-B2
	420	8500		350	298	230	90	5	38	8-Φ 22	120	25 x 14	100	
	150	738	F16	8.27	6.5	5.12	1.969	0.197	1.063	4-Φ0.866	2.756	0.551 x 0.354	2.244	GE-A5
	20	1000		210	165	130	50	5	27	4-Φ 22	70	14 x 9	57	
	300	1475	F16	8.27	6.5	5.12	1.969	0.197	1.063	4-Φ0.866	2.756	0.551 x 0.354	2.244	GE-A5
	50	2000		210	165	130	50	5	27	4-Φ 22	70	14 x 9	57	
	600	2581	F25	11.81	10	7.87	2.559	0.197	1.260	8-Φ0.709	3.543	0.709 x 0.433	2.874	GE-A8
	100	3500		300	254	200	65	5	32	8-Φ 18	90	18 x 11	73	
12 300	900	4057	F25	11.81	10	7.87	2.756	0.197	1.417	8-Φ0.709	3.937	0.787 x 0.472	3.11	GE-A9
	150	5500		300	254	200	70	5	36	8-Φ 18	100	20 x 12	79	
	1500	7744	F30	13.78	11.73	9.06	3.543	0.197	1.496	8-Φ0.866	4.724	0.984 x 0.551	3.937	GE-B3
	250	10500		350	298	230	90	5	38	8-Φ 22	120	25 x 14	100	
	2500	16226	F35	16.34	14.02	10.24	3.937	0.197	1.575	8-Φ 1.299	5.512	1.102 x 0.63	4.409	GE-B5
	420	22000		415	356	260	100	5	40	8-Φ 33	140	28 x 16	112	
	150	1106	F25	11.81	10	7.87	2.362	0.197	1.378	8-Φ0.709	3.15	0.709 x 0.433	2.677	GE-A7
	20	1500		300	254	200	60	5	35	8-Φ 18	80	18 x 11	68	
	300	2065	F25	11.81	10	7.87	2.362	0.197	1.378	8-Φ0.709	3.15	0.709 x 0.433	2.677	GE-A8
	50	2800		300	254	200	60	5	35	8-Φ 18	80	18 x 11	68	
12 300	600	4057	F25	11.81	10	7.87	2.756	0.197	1.378	8-Φ0.709	3.937	0.787 x 0.472	3.11	GE-A9
	100	5500		300	254	200	70	5	35	8-Φ 18	100	20 x 12	79	
	900	6491	F30	13.78	11.73	9.06	3.543	0.197	1.496	8-Φ0.866	4.724	0.984 x 0.551	3.937	GE-B2
	150	8800		350	298	230	90	5	38	8-Φ 22	120	25 x 14	100	
	1500	11801	F35	16.34	10.24	14.02	3.937	0.197	1.575	8-Φ 1.299	5.512	1.102 x 0.63	4.409	GE-B4
	250	16000		415	260	356	100	5	40	8-Φ 33	140	28 x 16	112	
	2500	18070	F35	16.34	10.24	1402	3.937	0.197	1.575	8-Φ 1.299	5.512	1.102 x 0.63	4.409	GE-C3
	420	24500		415	260	356	100	5	40	8-Φ 33	140	28 x 16	112	

Note:

(1) Torque doesn't include safety factor. The torque is based on seats of PTFE/RPTFE (Class 150 - 300), Nylon (Class 900 - 1500), and PEEK (Class 2500).

Torque shall be as follows if other than the base seat material is selected:

PTFE / RPTFE	Nylon	Devlon	PCTFE	TFM	Delrin	Molon	PPL	PEEK	Metal
Base	x 1	x 1	x 0.8	X 0.8	x 1	X 1	X 0.8	x 1.3	x 2.2

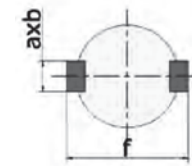
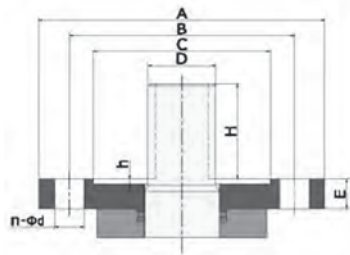
(2) Number of bolt holes and bolt hole diameter.



TRUNNION BALL VALVE

TOP MOUNTING DIMENSIONS AND TORQUE (CONT'D)

NPS 14-24 (DN 350-600)



Size in/mm	Rating Class/PN	Torque ⁽¹⁾ ft-lb/N·m	ISO F#	A	B	C	D	h	E	n-Φd ⁽²⁾	H	Drive Key a x b	f	Gearbox Model
14 350	150	1328	F25	11.81	10	7.87	2.559	0.197	1.299	8-Φ0.709	3.543	0.709 x 0.433	2.874	GE-A7
	20	1800		300	254	200	65	5	33	8-Φ18	90	18 x 11	73	
	300	2655	F25	11.81	10	7.87	2.559	0.197	1.299	8-Φ0.709	3.543	0.709 x 0.433	2.874	GE-A9
	50	3600		300	254	200	65	5	33	8-Φ18	90	18 x 11	73	
	600	5163	F25	11.81	10	7.87	3.15	0.197	1.378	8-Φ0.709	4.724	0.866 x 0.551	3.543	GE-B1
	100	7000		300	254	200	80	5	35	8-Φ18	120	22 x 14	90	
	900	8998	F30	13.78	11.73	9.06	3.937	0.197	1.378	8-Φ0.866	5.512	1.102 x 0.63	4.409	GE-B3
	150	12200		350	298	230	100	5	35	8-Φ22	140	28 x 16	112	
16 400	1500	13276	F35	16.34	14.02	10.24	3.937	0.197	1.575	8-Φ1.299	5.512	1.102 x 0.63	4.409	GE-B5
	250	18000		415	356	260	100	5	40	8-Φ33	140	28 x 16	112	
	150	2508	F25	11.81	10	7.78	3.15	0.197	1.378	8-Φ0.709	4.724	0.866 x 0.551	3.543	GE-A8
	20	3400		300	254	200	80	5	35	8-Φ18	120	22 x 14	90	
	300	4720	F25	11.81	10	7.78	3.15	0.197	1.378	8-Φ0.709	4.724	0.866 x 0.551	3.543	GE-A9
	50	6400		300	254	200	80	5	35	8-Φ18	120	22 x 14	90	
	600	8113	F30	13.78	11.73	9.06	3.937	0.197	1.378	8-Φ0.866	5.512	1.102 x 0.63	4.409	GE-B3
	100	11000		350	298	230	100	5	35	8-Φ22	140	28 x 16	112	
18 450	900	11432	F35	16.34	14.02	10.24	3.937	0.197	1.575	8-Φ1.299	5.512	1.102 x 0.63	4.409	GE-B4
	150	15500		415	356	260	100	5	40	8-Φ33	140	28 x 16	112	
	1500	18439	F40	18.7	15.98	11.81	34.724	0.315	1.772	8-Φ1.535	6.299	1.26 x 0.709	5.276	GE-C5
	250	25000		475	406	300	120	8	45	8-Φ39	160	32 x 18	134	
	150	2950	F25	11.81	10	7.87	3.15	0.197	1.378	8-Φ0.709	4.724	0.866 x 0.551	3.543	GE-A9
	20	4000		300	254	200	80	5	35	8-Φ18	120	22 x 14	90	
	300	5532	F25	11.81	10	7.87	3.15	0.197	1.378	8-Φ0.709	4.724	0.866 x 0.551	3.543	GE-A9
	50	7500		300	254	200	80	5	35	8-Φ18	120	22 x 14	90	
20 500	600	11063	F35	16.34	14.02	10.24	3.937	0.197	1.575	8-Φ1.299	5.512	1.102 x 0.63	4.409	GE-B4
	100	15000		415	356	260	100	5	40	8-Φ33	140	28 x 16	112	
	900	16226	F35	16.34	14.02	10.24	4.724	0.197	1.575	8-Φ1.299	6.299	1.26 x 0.709	5.276	GE-B5
	150	22000		415	356	260	120	5	40	8-Φ33	160	32 x 18	134	
	1500	26552	F40	18.7	15.98	11.81	5.512	0.315	1.772	8-Φ1.535	6.693	1.417 x 0.787	6.142	GE-B6
	250	36000		475	406	300	140	8	45	8-Φ39	170	36 x 20	156	
	150	3319	F30	13.78	11.73	9.06	3.543	0.197	1.496	8-Φ0.866	4.724	0.984 x 0.551	3.937	GE-B2
	20	4500		350	298	230	90	5	38	8-Φ22	120	25 x 14	100	
24 600	300	7007	F30	13.78	11.73	9.06	3.543	0.197	1.496	8-Φ0.866	4.724	0.984 x 0.551	3.937	GE-B3
	50	9500		350	298	230	90	5	38	8-Φ22	120	25 x 14	100	
	600	14014	F35	16.34	14.02	10.24	3.937	0.197	1.575	8-Φ1.299	5.512	1.102 x 0.63	4.409	GE-B5
	100	19000		415	356	260	100	5	40	8-Φ33	140	28 x 16	112	
	900	22864	F40	18.7	15.98	11.81	4.724	0.315	1.772	8-Φ1.535	6.299	1.26 x 0.709	5.276	GE-B6
	150	31000		475	406	300	120	8	45	8-Φ39	160	32 x 18	134	
	1500	35624	F48	22.05	19.02	14.57	5.512	0.315	1.772	12-Φ1.535	6.693	1.417 x 0.787	6.142	GE-B7
	250	48300		560	483	370	140	8	45	12-Φ39	170	36 x 20	156	
24 600	150	5900	F35	16.34	14.02	10.24	3.937	0.197	1.575	8-Φ1.299	5.512	1.102 x 0.63	4.409	GE-B4
	20	8000		415	356	260	100	5	40	8-Φ33	140	28 x 16	112	
	300	12907	F35	16.34	14.02	10.24	3.937	0.197	1.575	8-Φ1.299	5.512	1.102 x 0.63	4.409	GE-B4
	50	17500		415	356	260	100	5	40	8-Φ33	140	28 x 16	112	
	600	23602	F40	18.7	15.98	11.81	4.724	0.315	1.772	8-Φ1.535	6.299	1.26 x 0.709	5.276	GE-B6
	100	32000		475	406	300	120	8	45	8-Φ39	160	32 x 18	134	
	900	36878	F48	22.05	19.02	14.57	5.512	0.315	1.772	12-Φ1.535	6.693	1.417 x 0.787	6.142	GE-B7
	150	50000		560	483	370	140	8	45	12-Φ39	170	36 x 20	156	
24 600	1500	66380	F60	27.01	23.74	18.5	6.299	0.315	2.362	20-Φ1.535	7.48	1.575 x 0.866	7.008	GE-C7
	250	90000		686	603	470	160	8	60	20-Φ39	190	40 x 22	178	

Note:

(1) Torque doesn't include safety factor. The torque is based on seats of PTFE/RPTFE (Class 150 - 300), Nylon (Class 900 - 1500), and PEEK (Class 2500).

Torque shall be as follows if other than the base seat material is selected:

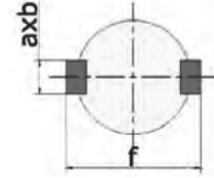
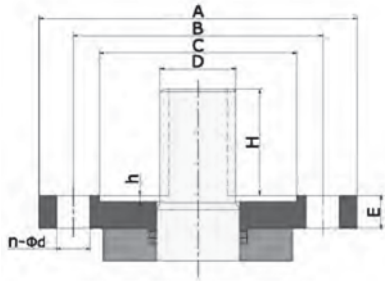
PTFE / RPTFE	Nylon	Devlon	PCTFE	TFM	Delrin	Malon	PPL	PEEK	Metal
Base	x 1	x 1	x 0.8	X 0.8	x 1	X 1	X 0.8	x 1.3	x 2.2

(2) Number of bolt holes and bolt hole diameter.

TRUNNION BALL VALVE

TOP MOUNTING DIMENSIONS AND TORQUE (CONT'D)

NPS 26-40 (DN 650-1000)



Size in/mm	Rating Class/PN	Torque ⁽¹⁾ ft-lb/N-m	ISO F#	A	B	C	D	h	E	n-Φd ⁽²⁾	H	Drive Key a x b	f	Gearbox Model
26 650	150	6269	F35	16.34	14.02	10.24	3.937	0.197	1.575	8-Φ 1.299	5.512	1.102 x 0.63	4.409	GE-B4
	20	8500		415	356	260	100	5	40	8-Φ 33	140	28 x 16	112	
	300	13276	F35	16.34	14.02	10.24	3.937	0.197	1.575	8-Φ 1.299	5.512	1.102 x 0.63	4.409	GE-B5
	50	18000		415	356	260	100	5	40	8-Φ 33	140	28 x 16	112	
	600	25077	F40	18.7	15.98	11.81	4.724	0.315	1.772	8-Φ 1.535	6.299	1.26 x 0.709	5.276	GE-B6
	100	34000		475	406	300	120	8	45	8-Φ 39	160	32 x 18	134	
28 700	900	42041	F48	22.05	19.02	14.57	5.512	0.315	1.772	12-Φ 1.535	6.693	1.417 x 0.787	6.142	GE-B7
	150	57000		560	483	370	140	8	45	12-Φ 39	170	36 x 20	156	
	150	6638	F35	16.34	14.02	10.24	3.937	0.197	1.575	8-Φ 1.299	5.512	1.102 x 0.63	4.409	GE-B4
	20	9000		415	356	260	100	5	40	8-Φ 33	140	28 x 16	112	
	300	16226	F35	16.34	14.02	10.24	3.937	0.197	1.575	8-Φ 1.299	5.512	1.102 x 0.63	4.409	GE-B5
	50	22000		415	356	260	100	5	40	8-Φ 33	140	28 x 16	112	
30 750	600	30978	F48	22.05	19.02	14.57	5.512	0.315	1.772	12-Φ 1.535	6.693	1.417 x 0.787	6.142	GE-B7
	100	42000		560	483	370	140	8	45	12-Φ 39	170	36 x 20	156	
	900	46465	F48	22.05	19.02	14.57	5.512	0.315	1.772	12-Φ 1.535	6.693	1.417 x 0.787	6.142	GE-B7
	150	63000		560	483	370	140	8	45	12-Φ 39	170	36 x 20	156	
	150	8851	F35	16.34	14.02	10.24	3.937	0.197	1.575	8-Φ 1.299	5.512	1.102 x 0.63	4.409	GE-B4
	20	12000		415	356	260	100	5	40	8-Φ 33	140	28 x 16	112	
32 800	300	19914	F40	18.7	15.98	11.81	4.724	0.315	1.772	8-Φ 1.535	6.299	1.26 x 0.709	5.276	GE-B6
	50	27000		475	406	300	120	8	45	8-Φ 39	160	32 x 18	134	
	600	37616	F48	22.05	19.02	14.57	5.512	0.315	1.772	12-Φ 1.535	6.693	1.417 x 0.787	6.142	GE-C5
	100	51000		560	483	370	140	8	45	12-Φ 39	170	36 x 20	156	
	900	55317	F48	22.05	19.02	14.57	6.299	0.315	1.969	12-Φ 1.535	7.48	1.575 x 0.866	7.008	GE-B7
	150	75000		560	483	370	160	8	50	12-Φ 39	190	40 x 22	178	
34 850	150	9957	F35	16.34	14.02	10.24	4.724	0.197	1.575	8-Φ 1.299	6.299	1.26 x 0.709	5.276	GE-B5
	20	13500		415	356	260	120	5	40	8-Φ 33	160	32 x 18	134	
	300	22864	F40	18.7	15.98	11.81	4.724	0.315	1.772	8-Φ 1.535	6.299	1.26 x 0.709	5.276	GE-B6
	50	31000		475	406	300	120	8	45	8-Φ 39	160	32 x 18	134	
	600	44254	F48	22.05	19.02	14.57	5.512	0.315	1.772	12-Φ 1.535	6.693	1.417 x 0.787	6.142	GE-C5
	100	60000		560	483	370	140	8	45	12-Φ 39	170	36 x 20	156	
36 900	150	11432	F35	16.34	14.02	10.24	4.724	0.197	1.575	8-Φ 1.299	6.299	1.26 x 0.709	5.276	GE-B5
	20	15500		415	356	260	120	5	40	8-Φ 33	160	32 x 18	134	
	300	26921	F40	18.7	15.98	11.81	5.512	0.315	1.772	8-Φ 1.535	6.693	1.417 x 0.787	6.142	GE-B6
	50	36500		475	406	300	140	8	45	8-Φ 39	170	36 x 20	156	
	600	51629	F48	22.05	19.02	14.57	6.299	0.315	1.969	12-Φ 1.535	7.48	1.575 x 0.866	7.008	GE-B7
	100	70000		560	483	370	160	8	50	12-Φ 39	190	40 x 22	178	
40 1000	150	12539	F35	16.34	14.02	10.24	4.724	0.197	1.575	8-Φ 1.299	6.299	1.26 x 0.709	5.276	GE-B5
	20	17000		415	356	260	120	5	40	8-Φ 33	160	32 x 18	134	
	300	28765	F40	18.7	15.98	11.81	5.512	0.315	1.772	8-Φ 1.535	6.693	1.417 x 0.787	6.142	GE-B6
	50	39000		475	406	300	140	8	45	8-Φ 39	170	36 x 20	156	
	600	56055	F48	22.05	19.02	14.57	6.299	0.315	1.969	12-Φ 1.535	7.48	1.575 x 0.866	7.008	GE-B7
	100	76000		560	483	370	160	8	50	12-Φ 39	190	40 x 22	178	
40 1000	150	18439	F40	18.7	15.98	11.81	7.724	0.315	1.772	8-Φ 1.535	6.299	1.26 x 0.709	5.276	GE-B6
	20	25000		475	406	300	120	8	45	8-Φ 39	160	32 x 18	134	
	300	36878	F48	22.05	19.02	14.57	5.512	0.315	1.969	12-Φ 1.535	6.693	1.417 x 0.787	6.142	GE-C5
	50	50000		560	483	370	140	8	50	12-Φ 39	170	36 x 20	156	
	600	66380	F60	27.01	23.74	18.5	6.299	0.315	2.362	20-Φ 1.535	7.48	1.575 x 0.866	7.008	GE-C7
	100	90000		686	603	470	160	8	60	20-Φ 39	190	40 x 22	178	

Note:

(1) Torque doesn't include safety factor. The torque is based on seats of PTFE/RPTFE (Class 150 - 300), Nylon (Class 900 - 1500), and PEEK (Class 2500).

Torque shall be as follows if other than the base seat material is selected:

PTFE / RPTFE	Nylon	Devlon	PCTFE	TFM	Delrin	Molon	PPL	PEEK	Metal
Base	x 1	x 1	x 0.8	X 0.8	x 1	X 1	X 0.8	x 1.3	x 2.2

(2) Number of bolt holes and bolt hole diameter.

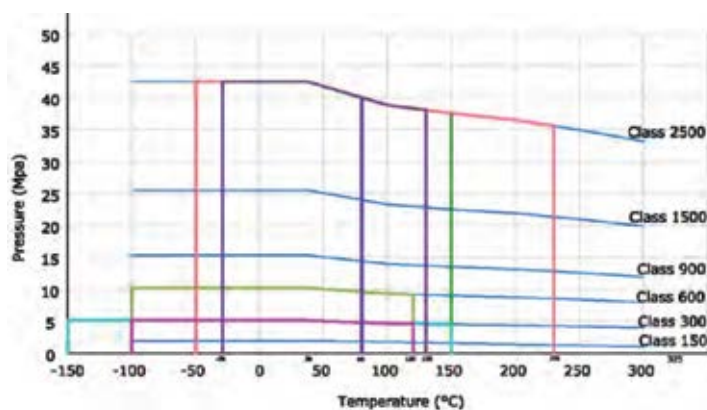


TRUNNION BALL VALVE

SEAT AND SEAL MATERIAL

SELECTION GUIDE

SEAT MATERIAL PRESSURE
TEMPERATURE RATING
(OPERATING)



SEAT MATERIALS

Material Name	Description	Operating Temperature	Operating Pressure	Service Application
PTFE	Virgin PTFE is the most widely used sealing material with excellent characteristics suitable for most services. It has excellent chemical resistance throughout valve industries and low coefficient of friction.	- 112 °F - 248 °F - 80 °C - 120 °C	Class 150 - 300 PN 20 - 50	General chemicals, low pressure services.
RP TFE	RPTFE (Reinforced PTFE) is typically produced by adding 1.5% fiber glass to virgin PTFE. It has better pressure - temperature properties than virgin PTFE, better resistance to wear and deformation under load. NOT to be used in hydrofluoroic acid	- 112 °F - 248 °F - 80 °C - 120 °C	Class 150 - 600 PN 20 - 100	For low and medium pressure services.
PCTFE	PCTFE is a homopolymer of chlorotrifluoroethylene, featuring high compressive strength and low deformation under load.	- 320 °F - 248 °F - 196 °C - 120 °C	Class 150 - 300 PN 20 - 50	For low temperature low pressure services.
Nylon 6	Nylon is a common seat material for Class 600 valves. It is highly resistance to many chemicals and abrasions, and can be used in air, oil and other gas media. It is NOT suitable for strong oxidization agents.	- 22 °F - 176 °F - 30 °C - 80 °C	Class 150 - 2500 PN 20 - 420	For high pressure, low temperature services.
Devlon®	Devlon® is a high molecular weight polyamide that is specifically tailored for high temperature/pressure applications in the offshore oil and gas sector. It is low moisture absorption.	- 50 °F - 302 °F - 46 °C - 150 °C	Class 150 - 2500 PN 20 - 420	For high pressure high temperature offshore services.
Nylon+MoS2	Molon (Nylon+MoS2) is a modified Nylon, the characteristics are similar to Devlon with it is cheaper than Devlon.	- 20 °F - 266 °F - 29 °C - 130 °C	Class 150 - 1500 PN 20 - 250	For high pressure, low temperature services.
PEEK	PEEK is a high performance engineered thermoplastic. It is excellent in water/chemical resistance and it is unaffected by continuous exposure to hot water/steam	- 148 °F - 500 °F - 100 °C - 260 °C	Class 600 - 2500 PN 100 - 420	For high pressure high temperature services.
PPL	PPL (Polyparaphenylene) is an excellent seat material with low coefficient of friction, highly resistant to pressure and temperature.	- 50 °F - 320 °F - 46 °C - 250 °C	Class 150 - 300 PN 20 - 50	For high temperature low pressure services.
Delrin®	Delrin® (Acetal Resin) possesses high tensile strength, creep resistance and toughness. It exhibits low moisture absorption. It is chemically resistant to hydrocarbons, solvents and neutral chemicals. DO NOT use it on oxygen service or steam.	- 50 °F - 194 °F - 46 °C - 90 °C	Class 150 - 600 PN 20 - 100	For extreme pressure services.
TFM	TFM (modified PTFE) is a chemically modified PTFE that offers enhanced properties while retaining all the proven advantages of a conventional PTFE.	- 112 °F - 320 °F - 80 °C - 160 °C	Class 150 - 600 PN 20 - 100	For services requiring high purity.
Metal	Metal (typically stellite) seats are used in severe conditions where flashing, hydraulic shock, abrasive media or trapped metal may exist in the line	Varies	Varies	For severe services.

O-RING MATERIALS

Material Name	Description	Operating Temperature	Operating Pressure
NBR	Buna - N (NBR) is an all purpose polymer with good resistance to water, solvents, oil and hydraulic fluids.	- 50 °F - 176 °F - 46 °C - 80 °C	Class 150 - 600 PN 20 - 100
HNBR	HNBR (Hydrogenated NBR) has similar media stability to NBR but with significantly better heat and oxidation stability.	- 50 °F - 176 °F - 46 °C - 80 °C	Class 150 - 600 PN 20 - 100
Viton	Viton (fluorocarbon) is a fluorocarbon elastomer that is compatible with a broad range of chemicals. It performs well in mineral acids, salt solutions, chlorinated hydrocarbons and petroleum oils.	- 49 °F - 320 °F - 22 °C - 204 °C	Class 150 - 600 PN 20 - 100
EPDM	EPDM has good abrasion and tear resistance with excellent chemical resistance to a variety of acids and alkalines. It is susceptible to attack by oil, strong acids and strong alkalines and should not be used in compressed air lines.	- 50 °F - 302 °F - 46 °C - 150 °C	Class 150 - 600 PN 20 - 100
FVMQ	Fluorosilicone is a silicone polymer chain with fluorinated side - chains for improved oil and fuel resistance. The mechanical and physical properties are very similar to those of silicone.	- 50 °F - 320 °F - 46 °C - 177 °C	Class 150 - 600 PN 20 - 100
AFLAS®	AFLAS® is highly resistant to a wide range of chemicals	- 49 °F - 428 °F - 29 °C - 220 °C	Class 150 - 600 PN 20 - 100

TRUNNION BALL VALVE

FLOW COEFFICIENT

SPECIFICATION TABLE

Nominal Size in/mm	Class 150 PN 20		Class 300 PN 50		Class 600 PN 100		Class 900 PN 150		Class 1500 PN 250	
	Cv	Kv	Cv	Kv	Cv	Kv	Cv	Kv	Cv	Kv
1/2 15	25	21	25	18	20	17	16	14	16	14
3/4 20	56	48	56	40	4	3	34	29	34	29
1 25	95	81	95	69	64	54	55	47	55	47
1 1/2 40	308	262	308	223	308	262	165	140	165	140
2 50	500	425	430	361	370	315	320	272	320	272
3 80	1,360	1,156	1,100	983	1,020	867	920	782	820	697
4 100	2,500	2,125	2,000	1,806	1,850	1,573	1,760	1,496	1,600	1,360
6 150	4,060	3,451	4,056	2,933	3,410	2,899	4,300	3,655	4,150	3,528
8 200	8,090	6,877	7,700	5,845	6,730	5,721	8,475	7,204	8,010	6,809
10 250	13,510	11,484	13,090	9,761	11,120	9,452	14,160	12,036	13,220	11,237
12 300	20,440	17,374	19,830	14,768	17,440	14,824	21,200	18,020	18,800	15,980
14 350	25,050	21,293	23,770	18,099	22,010	18,709	26,700	22,695	24,180	20,553
16 400	34,200	29,070	32,595	24,710	29,980	25,483	36,600	31,110	33,150	28,178
18 450	44,430	37,766	43,200	32,101	39,520	33,592	49,000	41,650	45,703	38,848
20 500	57,665	49,015	55,380	41,663	50,450	42,883	64,600	54,910	60,750	51,638
22 550	70,080	59,568	70,080	50,633	68,900	58,565				
24 600	87,680	74,528	84,720	63,349	76,630	65,136				
28 700	120,000	102,000	115,350	86,700	107,510	91,384				
30 750	141,850	120,573	136,600	102,487	125,630	106,786				
32 800	160,390	136,332	152,200	115,882	140,900	119,765				
36 900	205,450	174,633	192,995	148,438	239,160	203,286				
40 1000	248,700	211,395	248,700	179,686	239,160	203,286				
42 1050	275,260	233,971	275,260	198,875	275,260	233,971				
48 1200	364,180	309,553	364,180	263,120	364,180	309,553				
56 1400	529,430	450,016	529,430	382,513	520,500	442,425				

CALCULATION OF FLOW COEFFICIENT

Flow coefficient Cv (Kv is the metric equivalent) is the rate of flow in gallon per minute with the pressure drop of 1 psi across the valve. The flow coefficients shown in the above table are determined with equations as follows:

For liquids:

$$Q_l = C_v (\Delta P / SG)^{1/2}$$

Where:

Q_l = Flow of liquid (gallon/minute)

ΔP = Pressure drop in psi (P₁-P₂)

SG = Specific gravity (1 for liquid)

For gases (non-critical):

$$Q_g = 61 \cdot C_v (P_2 \cdot P_1 / SG)^{1/2}$$

Where:

Q_g = Flow of gases (SFH at STP)

P₂ = Outlet pressure (psi)

P₁ = Inlet pressure (psi)

SG = Specific gravity (1 for gas)



TRUNNION BALL VALVE

VALVE FIGURE NUMBER

HOW TO ORDER

Nominal Size	Valve Type	Pressure Rating	End Conn.	Construction	Body Material	Trim Material	Seat or Insert	O-ring	Operation Mode
A	B	C	D	E	F	G	H	I	J
e.g. 1 6	B A	6	R	4	C 1	1 0	N	1	G

is a NPS 16 Class 600 trunnion ball valve, RF flanged, 3-piece split body, A105 body, SS316 trim, Nylon seat insert, Viton O-ring, gear operated.

A Nominal Size							
00	Special	02	2 (DN 50)	07	8 RB	19	20 RB
F1	3/8 (DN 10)	R2	2 RB	08	8 (DN 200)	20	20 (DN 500)
F2	1/2 (DN 15)	F6	2 1/2 (DN 65)	09	10 RB	21	22 RB
OR	1/2 RB	3R	2 1/2 RB	10	10 (DN 250)	22	22 (DN 550)
F3	3/4 (DN 20)	03	3 (DN 80)	11	12 RB	23	24 RB
R0	3/4 RB	R3	3 RB	12	12 (DN 300)	24	24 (DN 600)
01	1 (DN 25)	04	4 (DN 100)	13	14 RB	25	26 RB
R1	1 RB	R4	4 RB	14	14 (DN 350)	26	26 (DN 650)
F4	1 1/4 (DN 32)	05	5 (DN 125)	15	16 RB	27	28 RB
1R	1 1/4 RB	R5	5 RB	16	16 (DN 400)	28	28 (DN 700)
F5	1 1/2 (DN 40)	06	6 (DN 150)	17	18 RB	29	30 RB
2R	1 1/2 RB	R6	6 RB	18	18 (DN 450)	30	30 (DN 750)
							... More as such

A Valve Type		C Pressure Rating		D End Connection	
BA	Ball Valve	0	Special	7	Class 2500
		1	Class 150	9	Class 900
		3	Class 300	8	Class 800
		5	Class 1500	2	Class 125
		6	Class 600	4	Class 400
				X	Special
				R	RF Flanged
				J	RTJ Flanged
				F	FF Flanged
				T	Threaded
				B	Butt-Weld (BW)
				S	Socket-Weld (SW)
				W	Wafer
				L	Lug

E Construction		F Body Material			
0	Special	X0	Special	M9	A182 F22 CL 3
3	2PC Body Trunnion	C1	A105	M0	A217 WC9
4	3PC Body Trunnion	C2	A216 WCA	E1	A182 F5
6	Welded Body Trunnion	C4	A216 WCB	E2	A217 C5
		C6	A216 WCC	E4	A217 C6
		M1	A182 F1	E5	A182 F9
		M2	A217 WC1	E6	A217 C12
		M3	A182 F2	E7	A182 F91
		M4	A217 WC4	E8	A217 C12A
		M5	A182 F12 CL 2	L1	A350 LF1
		M6	A217 WC5	L2	A352 LCA
		M7	A182 F11 CL 2	L3	A350 LF2
		M8	A217 WC6	L4	A352 LC2
				L5	A350 LF3
				L6	A352 LC3
				L7	A350 LF5
				L8	A352 LCB
				L9	A350 LF6
				LA	A350 LF9
				LB	A352 LC9
				LD	A352 LCC
				S1	A182 F304
				S2	A351 CF8
				S3	A182 F304L
				S4	A351 CF3
				S5	A182 F316
				S6	A351 CF8M
				S7	A182 F316L
				S8	A351 CF3M
				S9	A182 F347
				S0	A351 C F8C
				D1	A182 F51
				D2	A995 4A
				D3	A182 F53
				D4	A995 5A
				D5	A182 F55
				D6	A995 6A

G Trim Material					
00	Special	25	F53/F53/F53	41	A105+Ni60/A105+Ni55/4140+ENP
01	F6a/F6a/410	26	A105+ENP/A105+ENP/F316	42	A105+Ni60/A105+Ni60/17-4PH
02	304/304/304	27	304+ENP/304+ENP/304+ENP	44	Inconel 625/Inconel 625/Inconel 625
09	Monel/Monel/Monel	28	Inconel 625/Inconel 625/F316	45	316+ENP/316+ENP/F316
10	316/316/316	29	A105+ENP/A105+ENP/4140+ENP	46	316+ENP/316+ENP/F316+ENP
13	Alloy 20/Alloy 20/Alloy 20	30	A105+ENP/A105+ENP/A105+ENP	47	316+Ni60/316+Ni55/17-4PH
19	A105+ENP/A105+ENP/F6a	31	316/316/17-4PH	48	316L+Ni60/316L+Ni55/17-4PH
20	304L/304L/304L	36	LF2+ENP/LF2+ENP/F6a	49	316+Ni60/316+Ni55/4140+ENP
21	316L/316L/316L	37	A105+TCC/A105+TCC/17-4PH	50	316L+Ni60/316L+Ni55/4140+ENP
22	F321/F321/F321	38	A105+TCC/A105+TCC/4140+ENP		
23	F55/F55/F55	39	A105+TCC/A105+TCC/F51		
24	F51/F51/F51	40	A105+ENP/A105+ENP/17-4PH		

TRUNNION BALL VALVE

VALVE FIGURE NUMBER (CONT'D)

HOW TO ORDER

Nominal Size	Valve Type	Pressure Rating	End Conn.	Construction	Body Material	Trim Material	Seat or Insert	O-ring	Operation Mode
A	B	C	D	E	F	G	H	I	J
e.g. 16	BA	6	R	4	C1	10	N	1	G

is a NPS 16 Class 600 trunnion ball valve, RF flanged, 3-piece split body, A105 body, SS316 trim, Nylon seat insert, Viton O-ring, gear operated.

H	Seat or Seat Insert	I	O-Ring	J	Operation
X	Special	M	Metal	0	Bare Stem
T	PTFE	O	Molon	1	Lever
R	RPTFE	E	EPDM	2	Gearbox
N	Nylon	L	PPL	3	Pneumatic
V	Viton	A	Delrin	4	Electric
Q	TFM			5	Gear w/ Chain
D	Devlon			6	Pneumatic -Hydraulic
P	PEEK			7	Solid Lever
C	PCTFE			8	Lever with locking device
H	HNBR			9	Solid lever with locking device

HOW THE FIGURE NUMBER SYSTEM WORKS

Introduction. Figure number system uses a code consisting 14 digits of letters and numbers to represent the specification of a valve of certain specification. Of 14 digits, they are separated into 10 groups identified by letters from A to J. Each group represents a parameter of a valve, together they contain almost all the essential parameters of the valve.

Uses. To use the figure number system to generate a code is easy. Under each group, the code is shown on the left while on the right is the meaning of the code. Start by selecting a code from group A, through group J. If the specification of the valve is not listed, select the code for "Special". The total length of the figure number shall be exactly 14 digits.

Cautions. It is advised that you have as detailed the specification as possible to generate a figure number, which means eliminating "Specials". If you don't have enough specification or information about the valve you are ordering, or you're not sure how to use the system to generate a figure number, contact one of our sales representatives for help.

Note: Flow-Tek reserves the right to make any modifications without notice.



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